model_2500 · All 11 actual policy observations · env 24 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 24
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 24
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 24
DR · 0.00–60.00 s

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

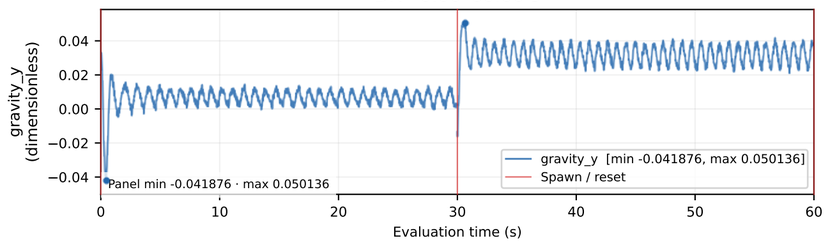
```
SAMPLED DR: env 24; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.02376, 1.12788, 0.816827, 1.00857]
  CoM XYZ mm [4.92334, 1.35443, -0.119776]
  mu_s [0.532645, 0.559809, 0.547743, 0.673378]
  mu_d [0.532645, 0.559809, 0.547743, 0.673378]
  Kp [steer,rear] [9.66888, 0]; Kd [0.192899, 0]
  spawn roll 0.0707435 rad; rate 0.193241 rad/s
  gyro bias XYZ rad/s [0.000166648, 0.000478024, 0.00031368]
  gravity bias YZ [3.62278e-05, 0.005236]
E2 [30.00, 40.88) s:
  mass x [1.05178, 0.919355, 0.940272, 1.03161]
  CoM XYZ mm [-4.1834, -1.93626, -2.92845]
  mu_s [0.547743, 0.592046, 0.632409, 0.576972]
  mu_d [0.547743, 0.534947, 0.589076, 0.576972]
  Kp [steer,rear] [10.494, 0]; Kd [0.192404, 0]
  spawn roll -0.0747468 rad; rate -0.16735 rad/s
  gyro bias XYZ rad/s [-3.86446e-05, -0.000305564, -0.000323233]
  gravity bias YZ [0.00139849, -0.000392816]
E3 [40.88, 60.00) s:
  mass x [0.966667, 0.967923, 1.07066, 1.15713]
  CoM XYZ mm [-4.86504, -4.15821, 3.49082]
  mu_s [0.794338, 0.632409, 0.703711, 0.602694]
  mu_d [0.526209, 0.589076, 0.703711, 0.602694]
  Kp [steer,rear] [9.02357, 0]; Kd [0.183519, 0]
  spawn roll 0.0516136 rad; rate -0.164886 rad/s
  gyro bias XYZ rad/s [-5.06925e-05, -6.31963e-05, 0.000165454]
  gravity bias YZ [0.00173651, 0.000749239]
Six significant digits shown; full precision: sampled_dr_values.json.
```

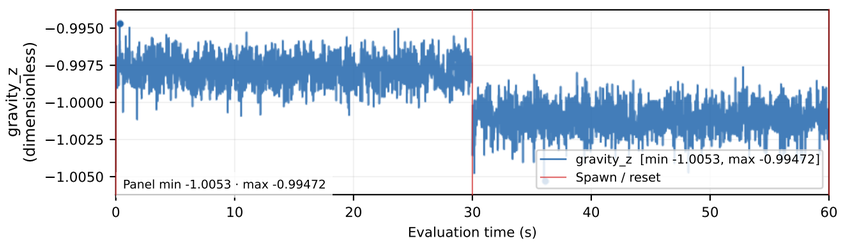
```
SAMPLED DR: env 24; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16949, 1.07713, 0.829174, 1.0497]
  CoM XYZ mm [-0.274809, 3.49019, -3.97172]
  mu_s [0.736077, 0.546287, 0.71155, 0.546287]
  mu_d [0.699455, 0.546287, 0.71155, 0.546287]
  Kp [steer,rear] [10.8176, 0]; Kd [0.201189, 0]
  spawn roll -0.00614922 rad; rate -0.118431 rad/s
  gyro bias XYZ rad/s [0.000194581, -0.000155248, 0.000214357]
  gravity bias YZ [-0.000313656, -0.000110502]
E2 [30.00, 30.22) s:
  mass x [1.11045, 1.01966, 1.1954, 0.860601]
  CoM XYZ mm [4.11354, -2.76793, 2.64046]
  mu_s [0.541939, 0.640867, 0.637675, 0.559972]
  mu_d [0.534359, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [9.4654, 0]; Kd [0.219437, 0]
  spawn roll 0.0981903 rad; rate -0.215889 rad/s
  gyro bias XYZ rad/s [0.000261415, 0.000154779, 0.000156588]
  gravity bias YZ [0.000304513, 0.000957138]
E3 [30.22, 30.66) s:
  mass x [1.03549, 1.0774, 0.809523, 0.982319]
  CoM XYZ mm [-2.63449, 0.558577, -3.9431]
  mu_s [0.728342, 0.568073, 0.616244, 0.640016]
  mu_d [0.728342, 0.568073, 0.586498, 0.513141]
  Kp [steer,rear] [9.28898, 0]; Kd [0.1849, 0]
  spawn roll -0.066071 rad; rate 0.0949529 rad/s
  gyro bias XYZ rad/s [-0.000316964, -0.000304982, -2.27981e-05]
  gravity bias YZ [-0.000705733, -0.00386866]
E4 [30.66, 60.00) s:
  mass x [1.00793, 0.966922, 0.803254, 1.08329]
  CoM XYZ mm [-3.46147, -0.00834489, -3.56986]
  mu_s [0.736077, 0.728342, 0.597748, 0.728342]
  mu_d [0.699455, 0.728342, 0.597748, 0.728342]
  Kp [steer,rear] [10.033, 0]; Kd [0.205122, 0]
  spawn roll -0.0216071 rad; rate 0.230526 rad/s
  gyro bias XYZ rad/s [0.00025827, 0.000150042, -1.7466e-05]
  gravity bias YZ [-0.00113656, 0.00450769]
Six significant digits shown; full precision: sampled_dr_values.json.
```

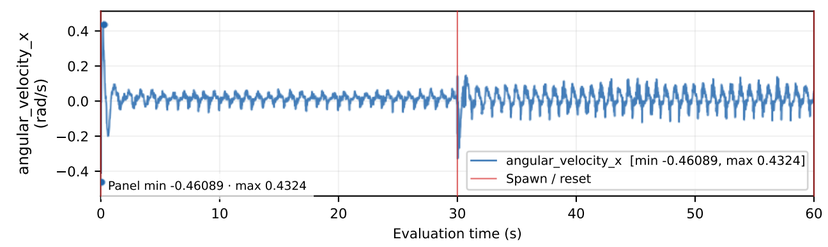
```
SAMPLED DR: env 24; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16949, 1.07713, 0.829174, 1.0497]
  CoM XYZ mm [-0.274809, 3.49019, -3.97172]
  mu_s [0.736077, 0.546287, 0.71155, 0.546287]
  mu_d [0.699455, 0.546287, 0.71155, 0.546287]
  Kp [steer,rear] [10.8176, 0]; Kd [0.201189, 0]
  spawn roll -0.00614922 rad; rate -0.118431 rad/s
  gyro bias XYZ rad/s [0.000194581, -0.000155248, 0.000214357]
  gravity bias YZ [-0.000313656, -0.000110502]
E2 [30.00, 30.18) s:
  mass x [1.11045, 1.01966, 1.1954, 0.860601]
  CoM XYZ mm [4.11354, -2.76793, 2.64046]
  mu_s [0.541939, 0.640867, 0.637675, 0.559972]
  mu_d [0.534359, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [9.4654, 0]; Kd [0.219437, 0]
  spawn roll 0.0981903 rad; rate -0.215889 rad/s
  gyro bias XYZ rad/s [0.000261415, 0.000154779, 0.000156588]
  gravity bias YZ [0.000304513, 0.000957138]
E3 [30.18, 31.58) s:
  mass x [1.1719, 1.00104, 1.09403, 1.01886]
  CoM XYZ mm [-2.9091, 3.2673, 1.6565]
  mu_s [0.791209, 0.728342, 0.637675, 0.528253]
  mu_d [0.767936, 0.728342, 0.637675, 0.528253]
  Kp [steer,rear] [10.0026, 0]; Kd [0.199406, 0]
  spawn roll -0.0513004 rad; rate 0.145146 rad/s
  gyro bias XYZ rad/s [-7.91602e-05, 0.000116882, 3.57925e-06]
  gravity bias YZ [0.000482736, 0.00138094]
E4 [31.58, 60.00) s:
  mass x [0.875317, 1.13192, 1.16229, 1.14476]
  CoM XYZ mm [-1.50508, 2.1826, 4.04661]
  mu_s [0.71008, 0.723429, 0.546287, 0.640016]
  mu_d [0.59371, 0.723429, 0.546287, 0.513141]
  Kp [steer,rear] [10.7536, 0]; Kd [0.215146, 0]
  spawn roll -0.0486227 rad; rate 0.215478 rad/s
  gyro bias XYZ rad/s [7.06148e-05, 5.34391e-06, 0.000158052]
  gravity bias YZ [0.00137741, -0.000929329]
Six significant digits shown; full precision: sampled_dr_values.json.
```

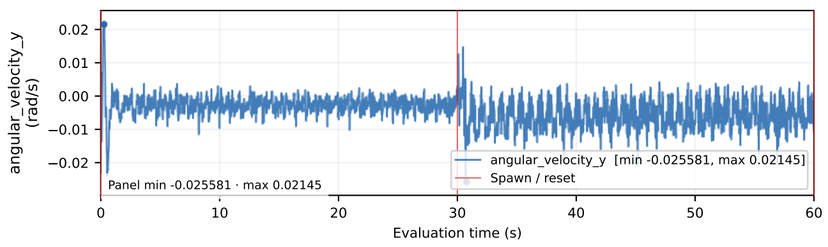
model_2500 · All 11 actual policy observations · env 25 · full 60 s

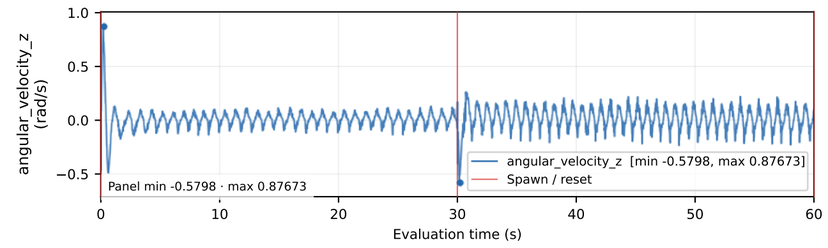
Isaac Lab · physics/PD 400 Hz · env 25
DR · 0.00–60.00 s

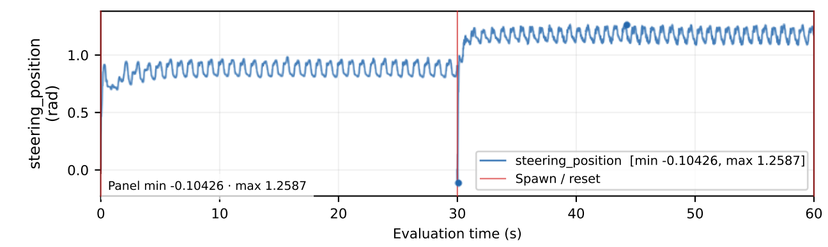


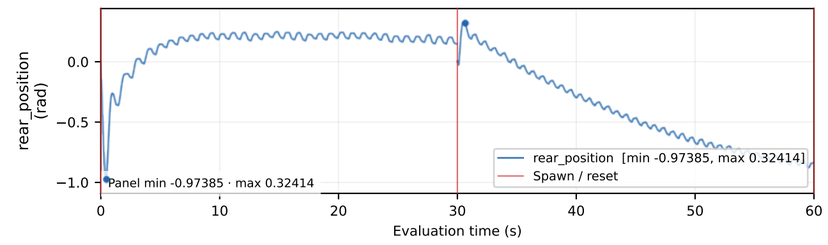


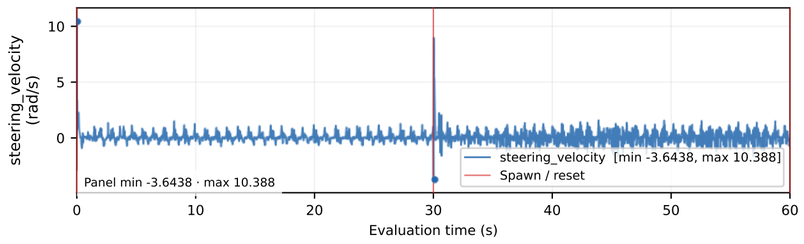


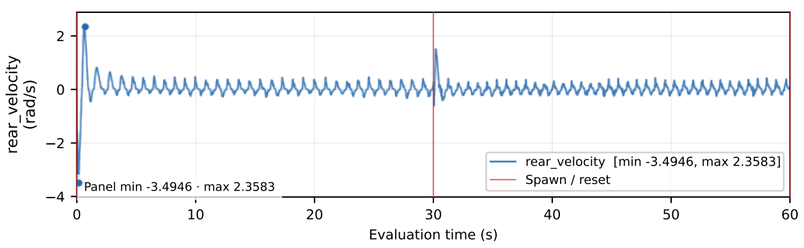


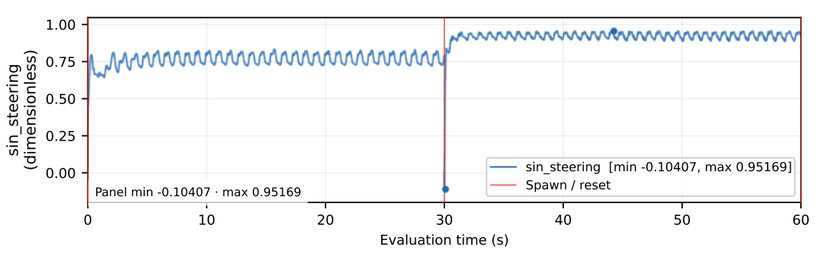


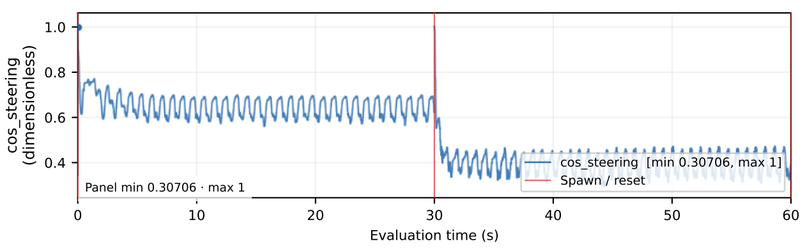




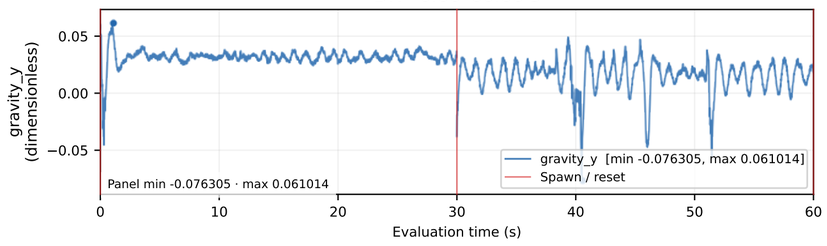


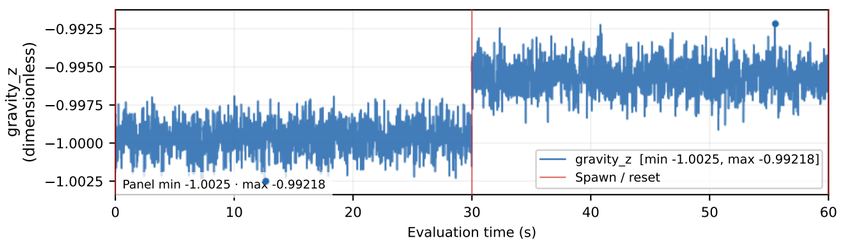


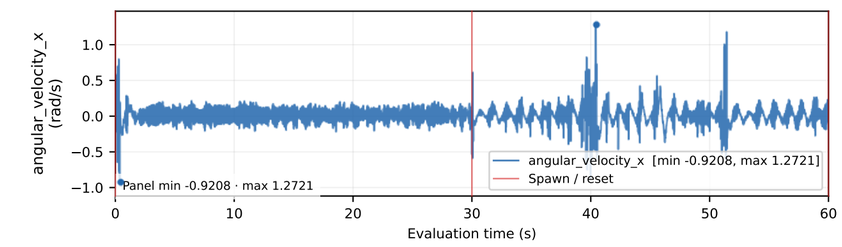


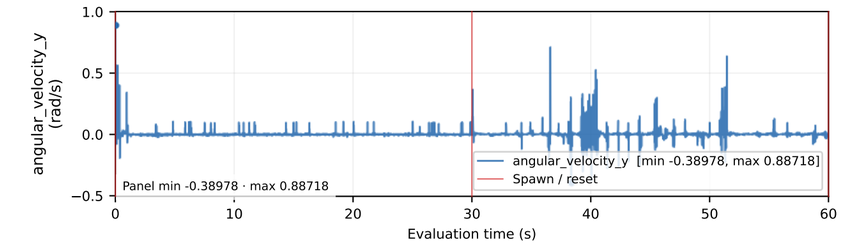


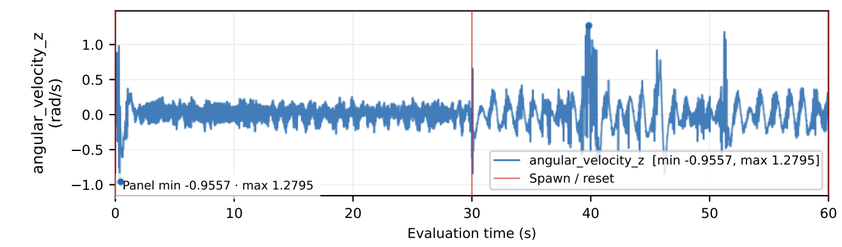
MuJoCo · physics/PD 400 Hz · env 25
DR · 0.00–60.00 s

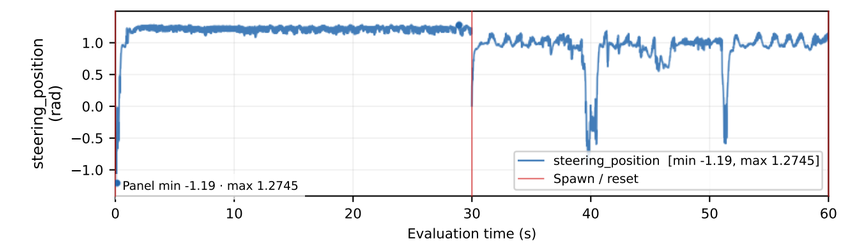


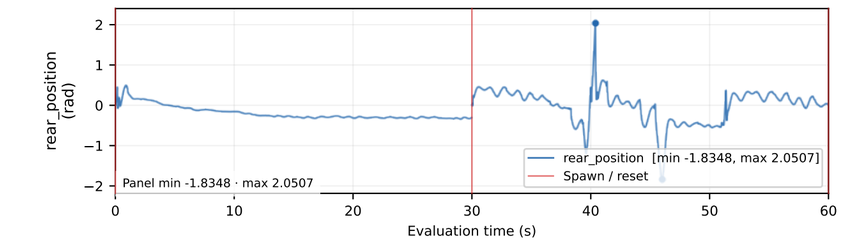


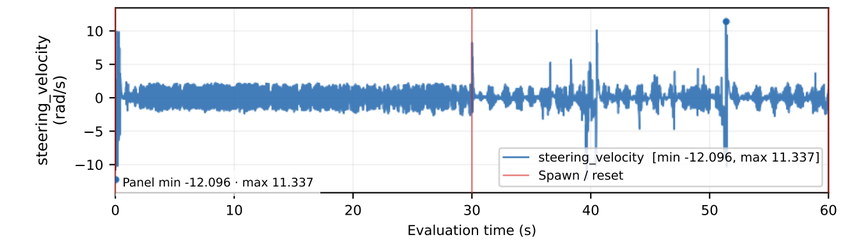


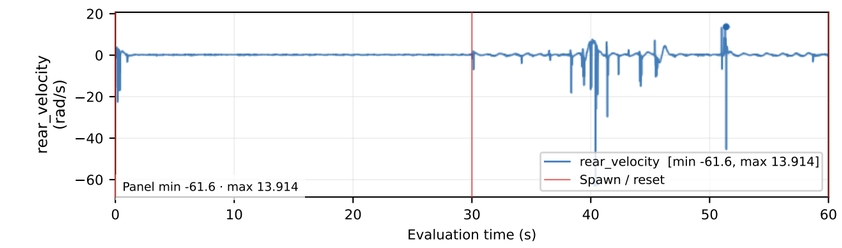


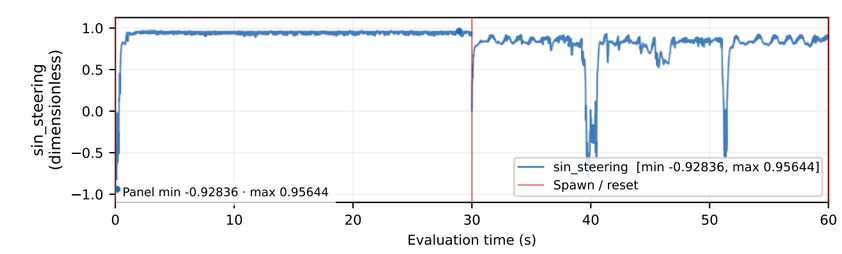


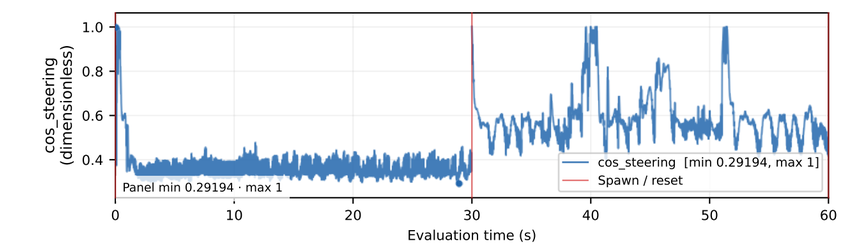




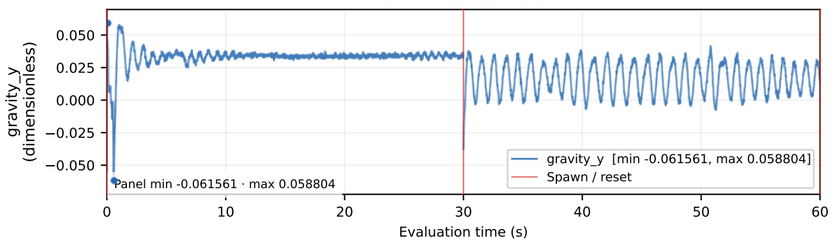


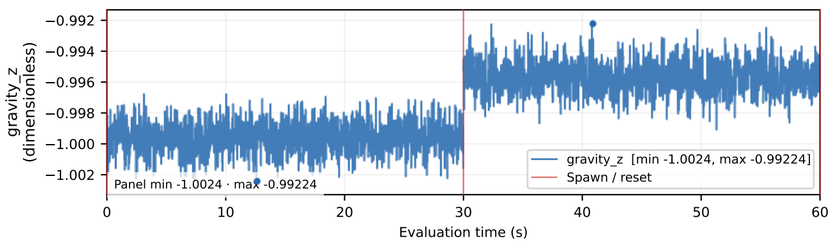


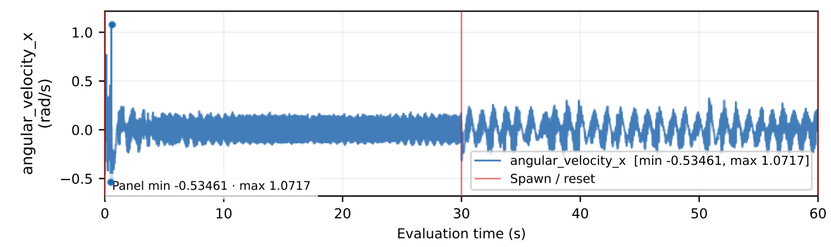


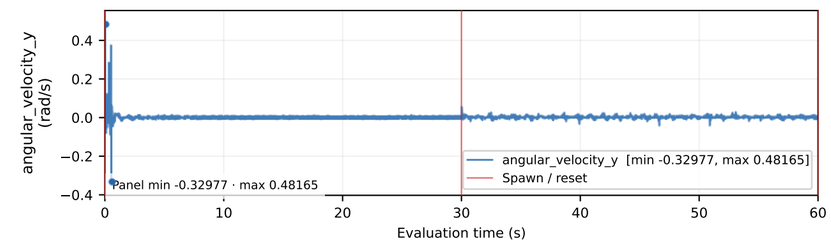


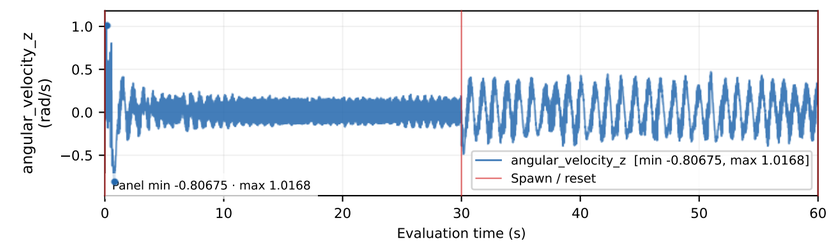
MuJoCo · physics/PD 800 Hz · env 25
DR · 0.00–60.00 s

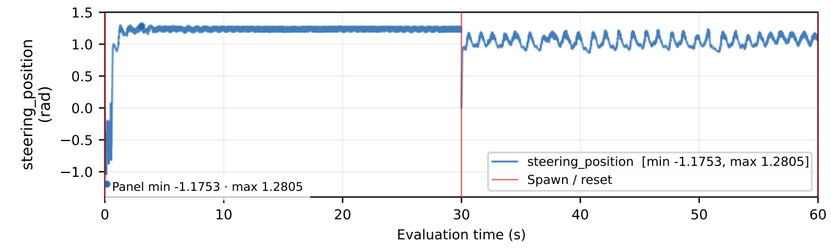


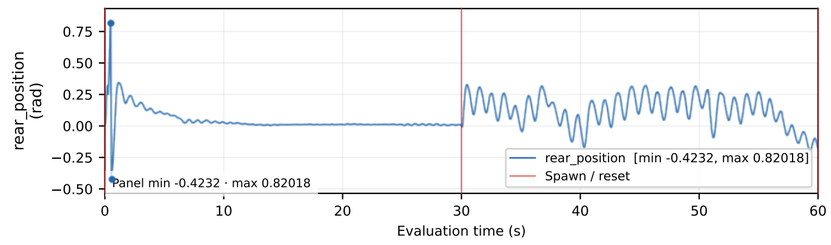


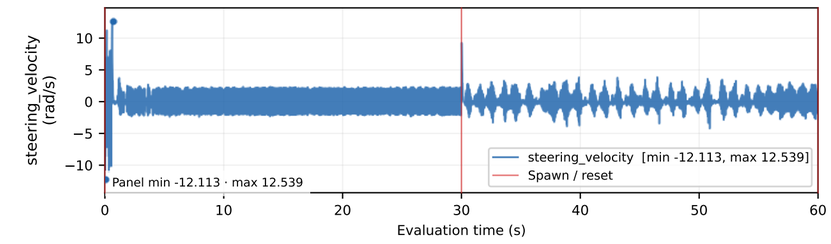


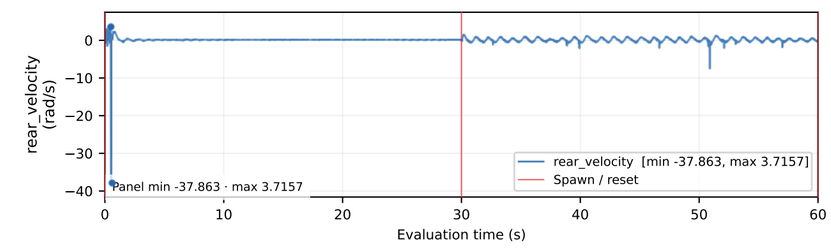


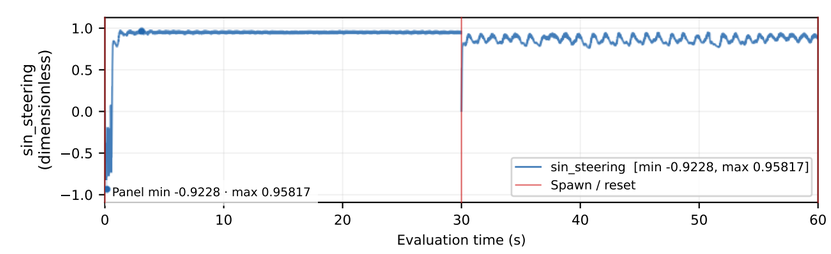


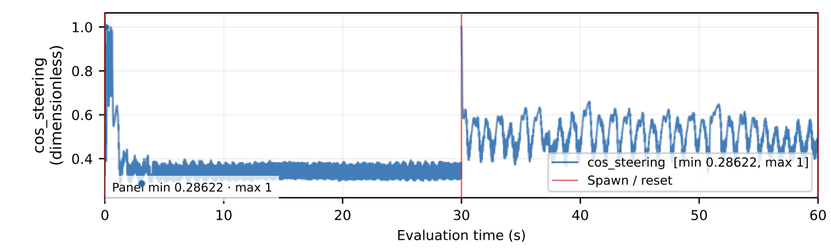












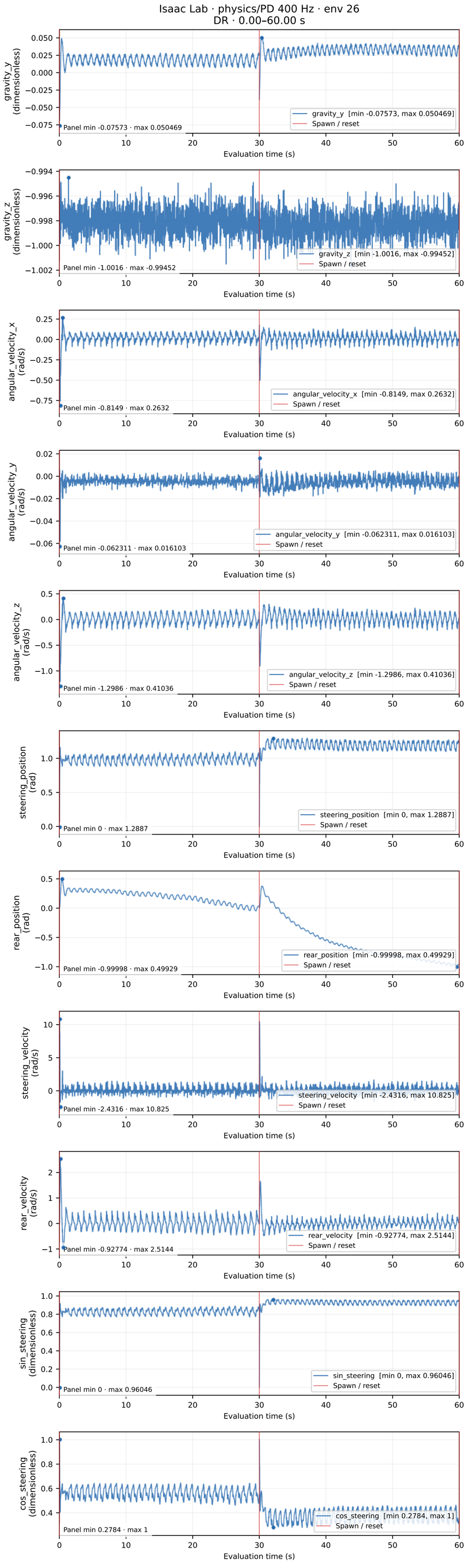
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

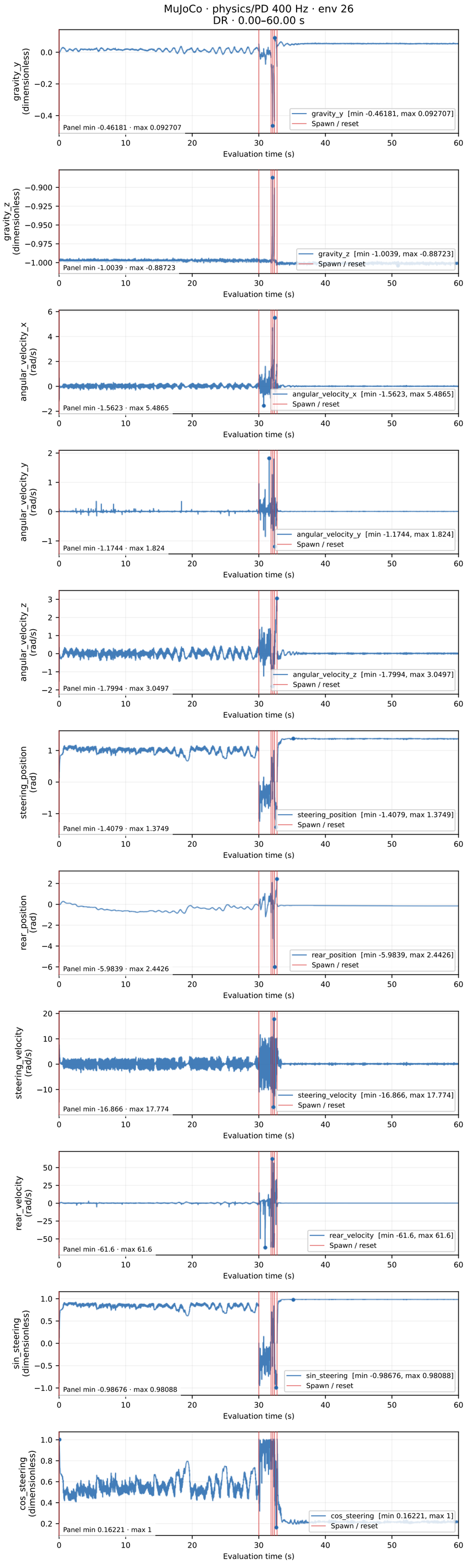
```
SAMPLED DR: env 25; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.875616, 1.17975, 0.903825, 0.929668]
  CoM XYZ mm [1.81167, 3.34438, -0.377148]
  mu_s [0.693205, 0.681711, 0.751971, 0.579974]
  mu_d [0.693205, 0.611756, 0.541224, 0.579974]
  Kp [steer,rear] [9.83792, 0]; Kd [0.197306, 0]
  spawn roll -0.0205786 rad; rate 0.173932 rad/s
  gyro bias XYZ rad/s [9.18702e-05, -4.10651e-05, -0.000261445]
  gravity bias YZ [0.00265045, 0.00207929]
E2 [30.00, 60.00) s:
  mass x [0.831212, 1.14263, 0.900257, 0.944867]
  CoM XYZ mm [0.970975, -0.505716, 1.01537]
  mu_s [0.531594, 0.741039, 0.647391, 0.741039]
  mu_d [0.531594, 0.614405, 0.647391, 0.614405]
  Kp [steer,rear] [9.08753, 0]; Kd [0.190724, 0]
  spawn roll 0.0144805 rad; rate -0.187457 rad/s
  gyro bias XYZ rad/s [-0.000286282, 0.000117795, -0.000100435]
  gravity bias YZ [0.000908414, -0.00157245]
Six significant digits shown; full precision: sampled_dr_values.json.
```

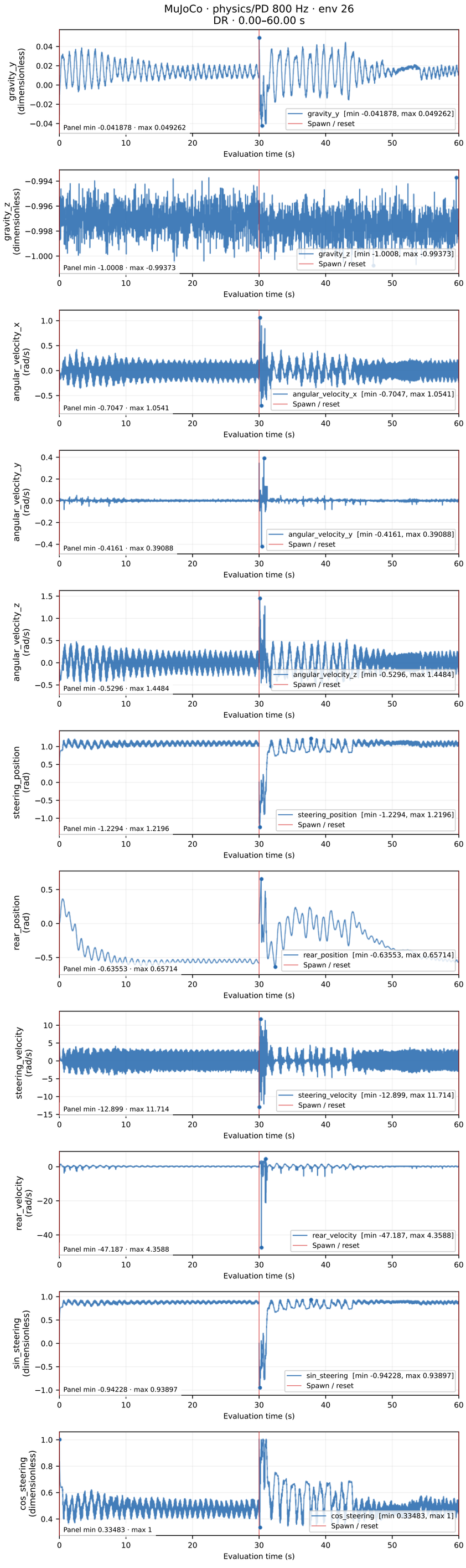
```
SAMPLED DR: env 25; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.965733, 0.850775, 1.00559, 0.918897]
  CoM XYZ mm [0.98037, -0.687294, -1.91706]
  mu_s [0.728342, 0.791209, 0.640016, 0.528253]
  mu_d [0.728342, 0.767936, 0.513141, 0.528253]
  Kp [steer,rear] [10.8245, 0]; Kd [0.180481, 0]
  spawn roll -0.0544291 rad; rate -0.0819372 rad/s
  gyro bias XYZ rad/s [-0.000260812, -0.000319122, -0.000211122]
  gravity bias YZ [0.00357963, -5.96259e-05]
E2 [30.00, 60.00) s:
  mass x [1.06177, 0.820528, 0.83899, 0.997132]
  CoM XYZ mm [1.44562, 1.0066, 0.635074]
  mu_s [0.546287, 0.616244, 0.631146, 0.669571]
  mu_d [0.546287, 0.586498, 0.631146, 0.669571]
  Kp [steer,rear] [9.37293, 0]; Kd [0.200151, 0]
  spawn roll 0.0338803 rad; rate 0.0723154 rad/s
  gyro bias XYZ rad/s [-0.000207155, 0.000191967, -0.000169671]
  gravity bias YZ [-0.00415054, 0.00403475]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 25; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.965733, 0.850775, 1.00559, 0.918897]
  CoM XYZ mm [0.98037, -0.687294, -1.91706]
  mu_s [0.728342, 0.791209, 0.640016, 0.528253]
  mu_d [0.728342, 0.767936, 0.513141, 0.528253]
  Kp [steer,rear] [10.8245, 0]; Kd [0.180481, 0]
  spawn roll -0.0544291 rad; rate -0.0819372 rad/s
  gyro bias XYZ rad/s [-0.000260812, -0.000319122, -0.000211122]
  gravity bias YZ [0.00357963, -5.96259e-05]
E2 [30.00, 60.00) s:
  mass x [1.06177, 0.820528, 0.83899, 0.997132]
  CoM XYZ mm [1.44562, 1.0066, 0.635074]
  mu_s [0.546287, 0.616244, 0.631146, 0.669571]
  mu_d [0.546287, 0.586498, 0.631146, 0.669571]
  Kp [steer,rear] [9.37293, 0]; Kd [0.200151, 0]
  spawn roll 0.0338803 rad; rate 0.0723154 rad/s
  gyro bias XYZ rad/s [-0.000207155, 0.000191967, -0.000169671]
  gravity bias YZ [-0.00415054, 0.00403475]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 26 · full 60 s







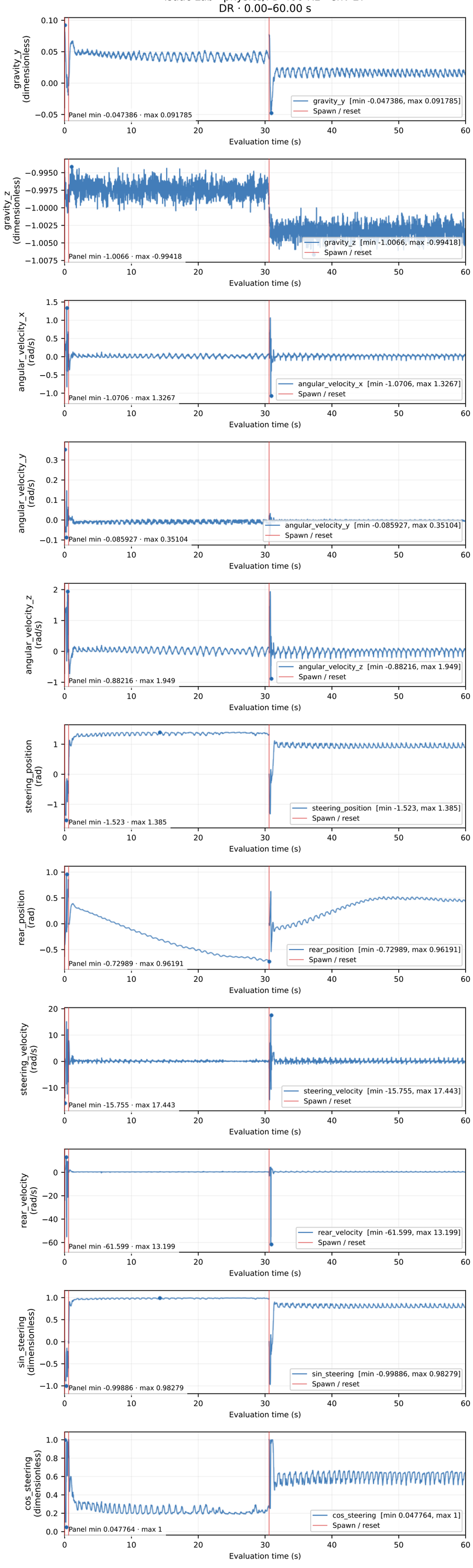
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

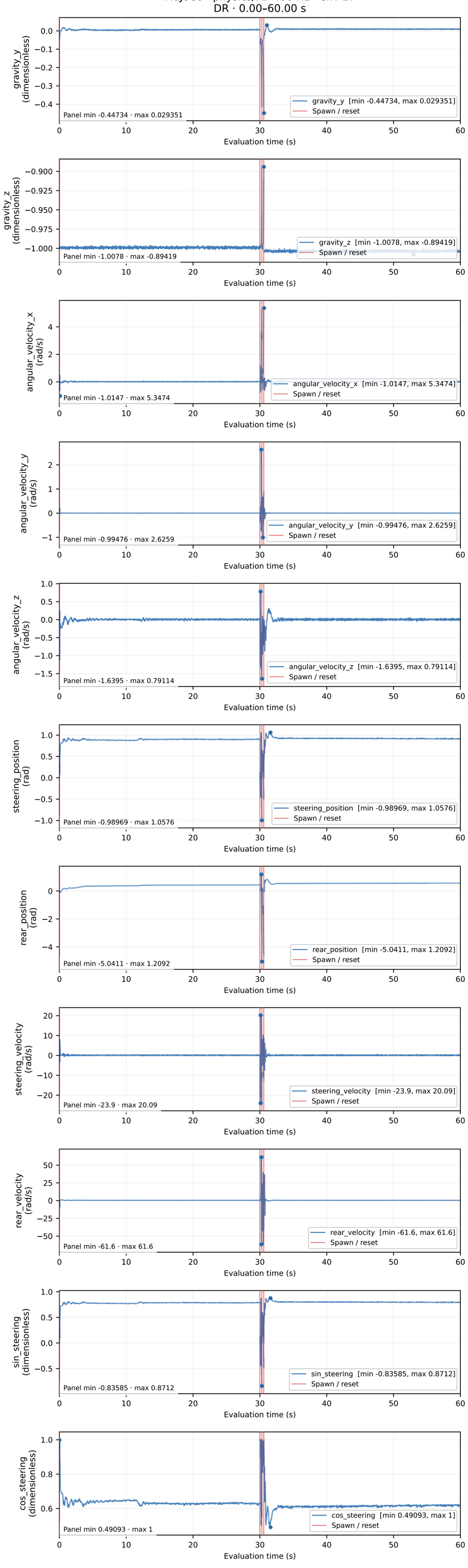
```
SAMPLED DR: env 26; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.915996, 0.906212, 1.03858, 1.1616]
  CoM XYZ mm [3.93371, 1.73644, -2.09413]
  mu_s [0.726175, 0.647391, 0.741039, 0.751971]
  mu_d [0.558574, 0.647391, 0.614405, 0.541224]
  Kp [steer,rear] [9.22716, 0]; Kd [0.204445, 0]
  spawn roll 0.0750011 rad; rate -0.0215228 rad/s
  gyro bias XYZ rad/s [0.000132369, -0.000562396, 1.03834e-05]
  gravity bias YZ [-0.000777999, 0.00171696]
E2 [30.00, 60.00) s:
  mass x [0.877824, 1.01535, 1.1085, 1.17473]
  CoM XYZ mm [-4.86545, -1.24921, -4.78084]
  mu_s [0.628821, 0.559809, 0.659445, 0.559809]
  mu_d [0.628821, 0.559809, 0.547619, 0.559809]
  Kp [steer,rear] [10.4904, 0]; Kd [0.211355, 0]
  spawn roll 0.0364383 rad; rate 0.174227 rad/s
  gyro bias XYZ rad/s [0.000113855, 0.000254793, -0.000328278]
  gravity bias YZ [-0.00144561, 0.00106124]
Six significant digits shown; full precision: sampled_dr_values.json.
```

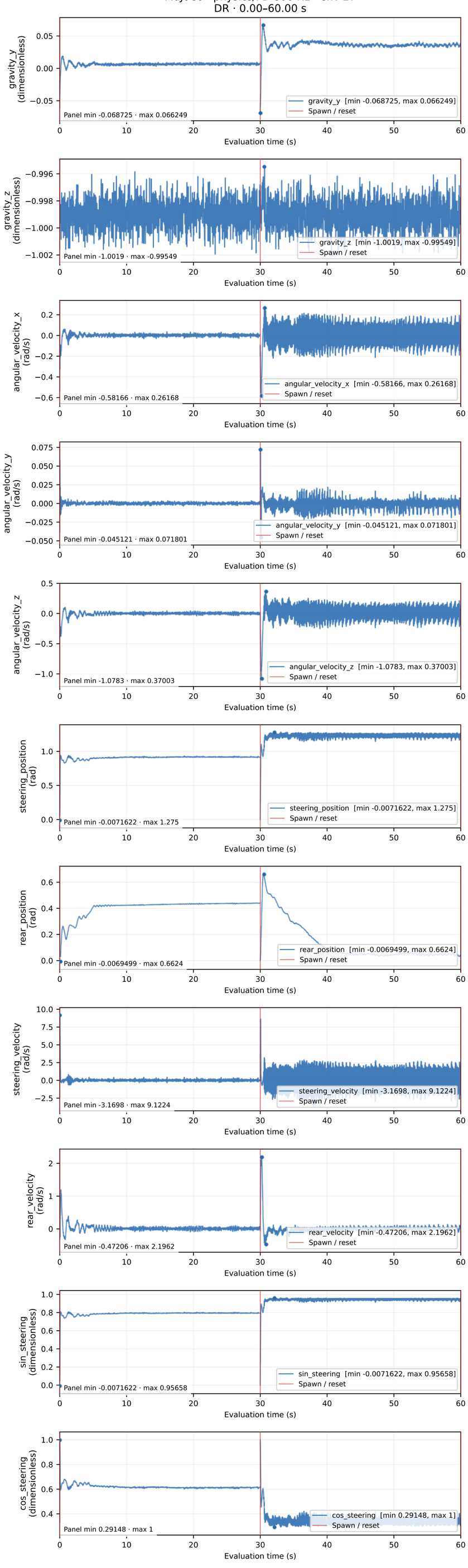
```
SAMPLED DR: env 26; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.993747, 1.13933, 1.06941, 1.0505]
  CoM XYZ mm [4.75818, 1.9279, -3.69763]
  mu_s [0.723429, 0.597748, 0.538976, 0.568073]
  mu_d [0.723429, 0.597748, 0.538976, 0.568073]
  Kp [steer,rear] [9.35545, 0]; Kd [0.209963, 0]
  spawn roll 0.0290254 rad; rate 0.209869 rad/s
  gyro bias XYZ rad/s [0.00013998, 0.000133799, -1.10574e-05]
  gravity bias YZ [-0.00157627, 0.00273748]
E2 [30.00, 31.84) s:
  mass x [0.885949, 1.08233, 0.917855, 1.03555]
  CoM XYZ mm [-1.94472, 1.7385, -4.32292]
  mu_s [0.669571, 0.669571, 0.757579, 0.748289]
  mu_d [0.669571, 0.669571, 0.70921, 0.689499]
  Kp [steer,rear] [10.2197, 0]; Kd [0.18084, 0]
  spawn roll -0.0491084 rad; rate -0.133235 rad/s
  gyro bias XYZ rad/s [2.47219e-05, -0.000191236, 2.73441e-05]
  gravity bias YZ [-0.0014935, 0.00221651]
E3 [31.84, 32.10) s:
  mass x [0.942888, 0.971694, 1.11327, 1.06906]
  CoM XYZ mm [0.537742, 0.536609, 0.630142]
  mu_s [0.611239, 0.631146, 0.728342, 0.528253]
  mu_d [0.611239, 0.631146, 0.728342, 0.528253]
  Kp [steer,rear] [9.96441, 0]; Kd [0.192504, 0]
  spawn roll 0.0934041 rad; rate -0.144448 rad/s
  gyro bias XYZ rad/s [0.000218819, -0.000101322, -0.000225819]
  gravity bias YZ [-0.000608424, -0.00229416]
E4 [32.10, 32.36) s:
  mass x [1.12464, 0.953291, 1.0491, 0.924926]
  CoM XYZ mm [0.0679545, -0.031636, 3.96472]
  mu_s [0.704749, 0.736077, 0.749678, 0.666375]
  mu_d [0.541926, 0.699455, 0.741429, 0.519145]
  Kp [steer,rear] [9.76565, 0]; Kd [0.200747, 0]
  spawn roll 0.0978547 rad; rate 0.198396 rad/s
  gyro bias XYZ rad/s [-1.95181e-05, 0.000111424, -0.00014902]
  gravity bias YZ [-0.00139232, -0.000730083]
E5 [32.36, 32.74) s:
  mass x [0.922887, 1.11895, 0.973031, 1.1487]
  CoM XYZ mm [4.99285, 2.89677, 3.46991]
  mu_s [0.633024, 0.631146, 0.748289, 0.732187]
  mu_d [0.568172, 0.631146, 0.689499, 0.631664]
  Kp [steer,rear] [10.1915, 0]; Kd [0.217585, 0]
  spawn roll -0.0739943 rad; rate -0.0990864 rad/s
  gyro bias XYZ rad/s [-0.000252783, -1.43901e-05, 0.000141257]
  gravity bias YZ [-0.000742643, -0.00254135]
E6 [32.74, 60.00) s:
  mass x [0.818999, 1.19359, 0.905628, 1.02562]
  CoM XYZ mm [-4.64729, -3.71685, -0.274436]
  mu_s [0.704749, 0.640867, 0.736077, 0.791209]
  mu_d [0.541926, 0.556841, 0.699455, 0.767936]
  Kp [steer,rear] [9.04605, 0]; Kd [0.1897, 0]
  spawn roll -0.0316723 rad; rate 0.158515 rad/s
  gyro bias XYZ rad/s [-6.09537e-05, -2.27372e-06, 0.000226784]
  gravity bias YZ [0.003926, -0.00216295]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 26; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.993747, 1.13933, 1.06941, 1.0505]
  CoM XYZ mm [4.75818, 1.9279, -3.69763]
  mu_s [0.723429, 0.597748, 0.538976, 0.568073]
  mu_d [0.723429, 0.597748, 0.538976, 0.568073]
  Kp [steer,rear] [9.35545, 0]; Kd [0.209963, 0]
  spawn roll 0.0290254 rad; rate 0.209869 rad/s
  gyro bias XYZ rad/s [0.00013998, 0.000133799, -1.10574e-05]
  gravity bias YZ [-0.00157627, 0.00273748]
E2 [30.00, 60.00) s:
  mass x [0.885949, 1.08233, 0.917855, 1.03555]
  CoM XYZ mm [-1.94472, 1.7385, -4.32292]
  mu_s [0.669571, 0.669571, 0.757579, 0.748289]
  mu_d [0.669571, 0.669571, 0.70921, 0.689499]
  Kp [steer,rear] [10.2197, 0]; Kd [0.18084, 0]
  spawn roll -0.0491084 rad; rate -0.133235 rad/s
  gyro bias XYZ rad/s [2.47219e-05, -0.000191236, 2.73441e-05]
  gravity bias YZ [-0.0014935, 0.00221651]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 27 · full 60 s







50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

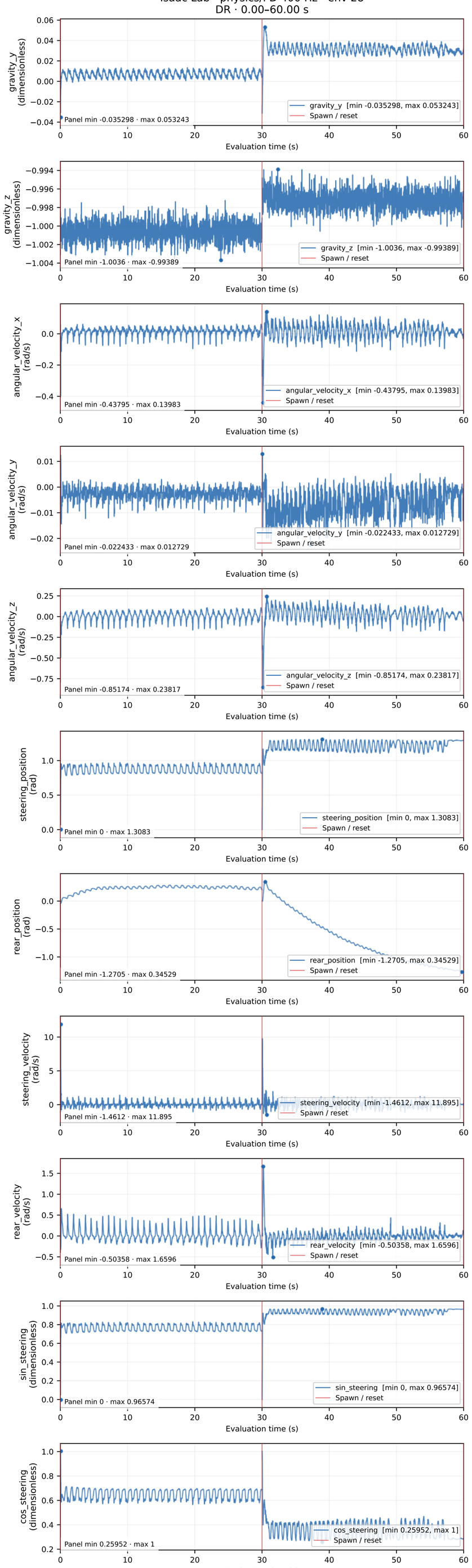
```
SAMPLED DR: env 27; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 0.60) s:
  mass x [1.14218, 1.17407, 0.880879, 0.815383]
  CoM XYZ mm [-3.5619, 4.63646, 2.62785]
  mu_s [0.539956, 0.533455, 0.760821, 0.774409]
  mu_d [0.539956, 0.533455, 0.670315, 0.561095]
  Kp [steer,rear] [10.978, 0]; Kd [0.180798, 0]
  spawn roll -0.0902905 rad; rate 0.206057 rad/s
  gyro bias XYZ rad/s [-9.91288e-05, -3.06844e-06, -2.71612e-05]
  gravity bias YZ [0.0023112, 0.00116169]
E2 [0.60, 30.60) s:
  mass x [1.09414, 0.88537, 0.903026, 0.999329]
  CoM XYZ mm [-1.14301, -3.40579, 4.10023]
  mu_s [0.751971, 0.751971, 0.751971, 0.547743]
  mu_d [0.541224, 0.541224, 0.541224, 0.547743]
  Kp [steer,rear] [9.60229, 0]; Kd [0.183213, 0]
  spawn roll 0.00356567 rad; rate -0.112306 rad/s
  gyro bias XYZ rad/s [6.14946e-06, 0.000212763, -0.000417311]
  gravity bias YZ [-0.00135359, 0.00155985]
E3 [30.60, 60.00) s:
  mass x [1.00297, 0.822399, 1.13368, 1.1895]
  CoM XYZ mm [-3.1565, 1.60713, 3.7401]
  mu_s [0.632409, 0.787792, 0.598343, 0.673378]
  mu_d [0.589076, 0.617134, 0.598343, 0.673378]
  Kp [steer,rear] [10.8456, 0]; Kd [0.217123, 0]
  spawn roll -0.0781943 rad; rate 0.0748008 rad/s
  gyro bias XYZ rad/s [0.000252128, -9.52991e-05, -5.25896e-05]
  gravity bias YZ [-0.000183114, -0.00352336]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 27; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.03215, 1.01074, 1.14598, 1.0939]
  CoM XYZ mm [-0.234152, 3.16696, 0.446348]
  mu_s [0.559972, 0.640016, 0.559972, 0.732187]
  mu_d [0.502209, 0.513141, 0.502209, 0.631664]
  Kp [steer,rear] [10.3569, 0]; Kd [0.208237, 0]
  spawn roll 0.0467654 rad; rate -0.215901 rad/s
  gyro bias XYZ rad/s [0.000142147, -3.23553e-05, -8.01081e-05]
  gravity bias YZ [0.00221784, 0.000939233]
E2 [30.00, 30.30) s:
  mass x [1.0161, 0.989662, 0.917133, 0.918384]
  CoM XYZ mm [3.42184, -0.646235, 1.00044]
  mu_s [0.666375, 0.723429, 0.727426, 0.597748]
  mu_d [0.519145, 0.723429, 0.606358, 0.597748]
  Kp [steer,rear] [9.98354, 0]; Kd [0.218973, 0]
  spawn roll 0.0708753 rad; rate -0.194923 rad/s
  gyro bias XYZ rad/s [6.70637e-05, 2.76081e-05, 0.000159323]
  gravity bias YZ [0.00398194, 0.000394695]
E3 [30.30, 30.58) s:
  mass x [0.964189, 1.19385, 1.16629, 1.00441]
  CoM XYZ mm [3.59202, -1.0898, -0.457886]
  mu_s [0.700521, 0.528253, 0.538976, 0.611239]
  mu_d [0.641329, 0.528253, 0.538976, 0.611239]
  Kp [steer,rear] [9.55067, 0]; Kd [0.214054, 0]
  spawn roll 0.0634218 rad; rate 0.0615261 rad/s
  gyro bias XYZ rad/s [0.000118771, -0.000156696, -7.15329e-05]
  gravity bias YZ [-0.00242644, -0.000123051]
E4 [30.58, 60.00) s:
  mass x [0.846888, 0.844771, 1.08345, 1.06525]
  CoM XYZ mm [-4.91461, 2.57639, -2.21418]
  mu_s [0.640867, 0.559972, 0.69316, 0.640016]
  mu_d [0.556841, 0.502209, 0.69316, 0.513141]
  Kp [steer,rear] [10.5368, 0]; Kd [0.184188, 0]
  spawn roll 0.0444598 rad; rate 0.101519 rad/s
  gyro bias XYZ rad/s [-0.000230281, -0.000219928, -0.000218251]
  gravity bias YZ [0.00296641, -0.00362709]
Six significant digits shown; full precision: sampled_dr_values.json.
```

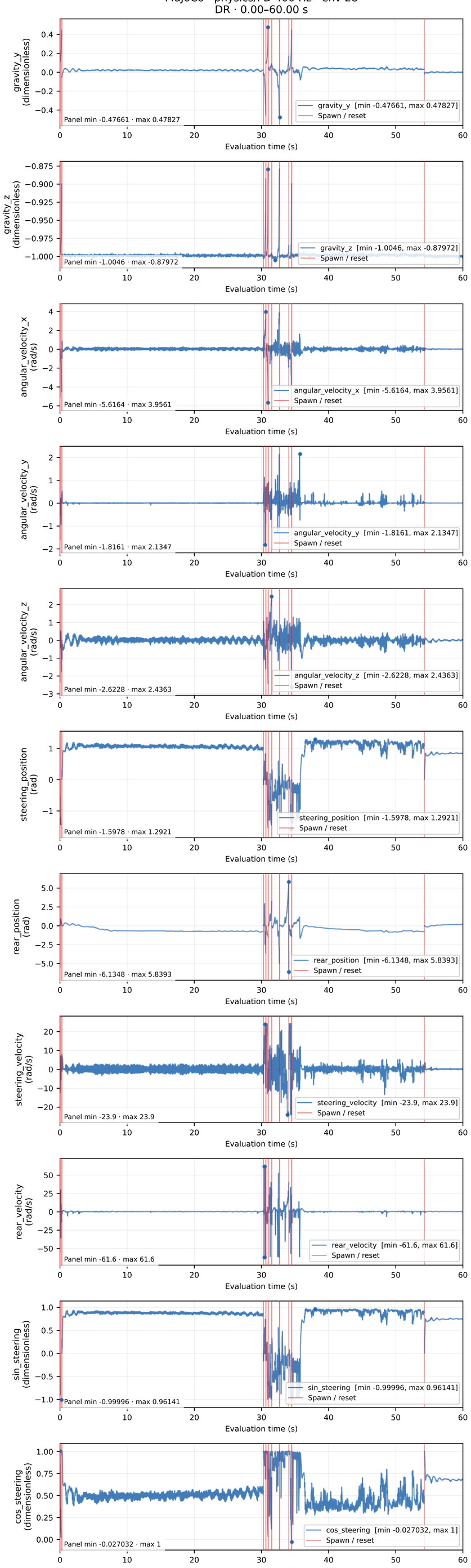
```
SAMPLED DR: env 27; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.03215, 1.01074, 1.14598, 1.0939]
  CoM XYZ mm [-0.234152, 3.16696, 0.446348]
  mu_s [0.559972, 0.640016, 0.559972, 0.732187]
  mu_d [0.502209, 0.513141, 0.502209, 0.631664]
  Kp [steer,rear] [10.3569, 0]; Kd [0.208237, 0]
  spawn roll 0.0467654 rad; rate -0.215901 rad/s
  gyro bias XYZ rad/s [0.000142147, -3.23553e-05, -8.01081e-05]
  gravity bias YZ [0.00221784, 0.000939233]
E2 [30.00, 60.00) s:
  mass x [1.0161, 0.989662, 0.917133, 0.918384]
  CoM XYZ mm [3.42184, -0.646235, 1.00044]
  mu_s [0.666375, 0.723429, 0.727426, 0.597748]
  mu_d [0.519145, 0.723429, 0.606358, 0.597748]
  Kp [steer,rear] [9.98354, 0]; Kd [0.218973, 0]
  spawn roll 0.0708753 rad; rate -0.194923 rad/s
  gyro bias XYZ rad/s [6.70637e-05, 2.76081e-05, 0.000159323]
  gravity bias YZ [0.00398194, 0.000394695]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 28 · full 60 s

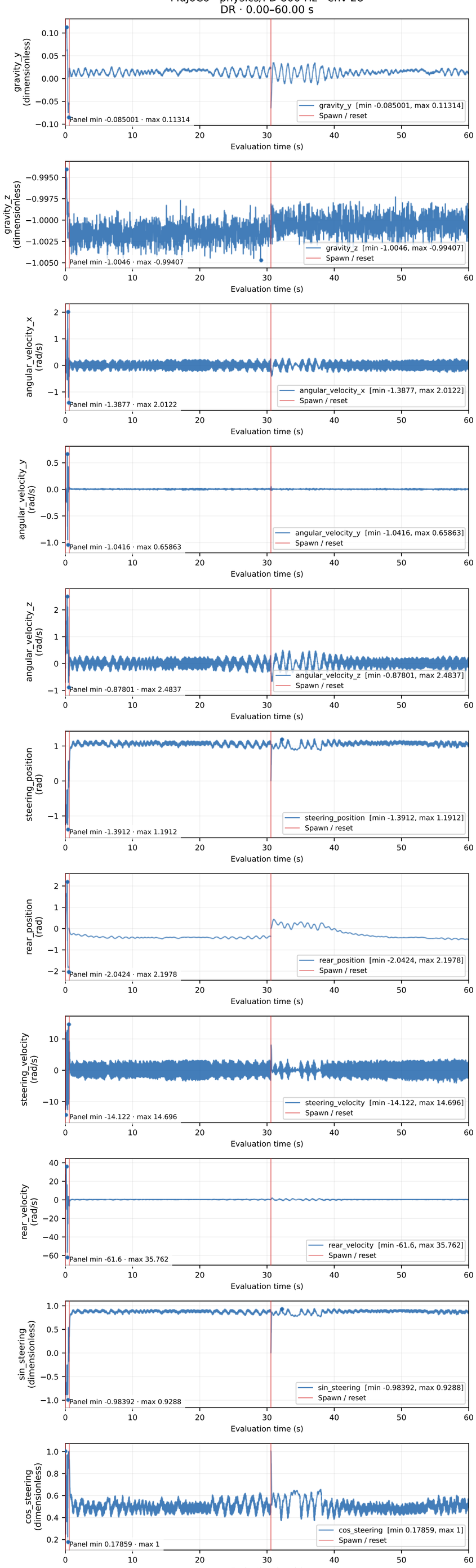
Isaac Lab · physics/PD 400 Hz · env 28
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 28
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 28
DR · 0.00–60.00 s



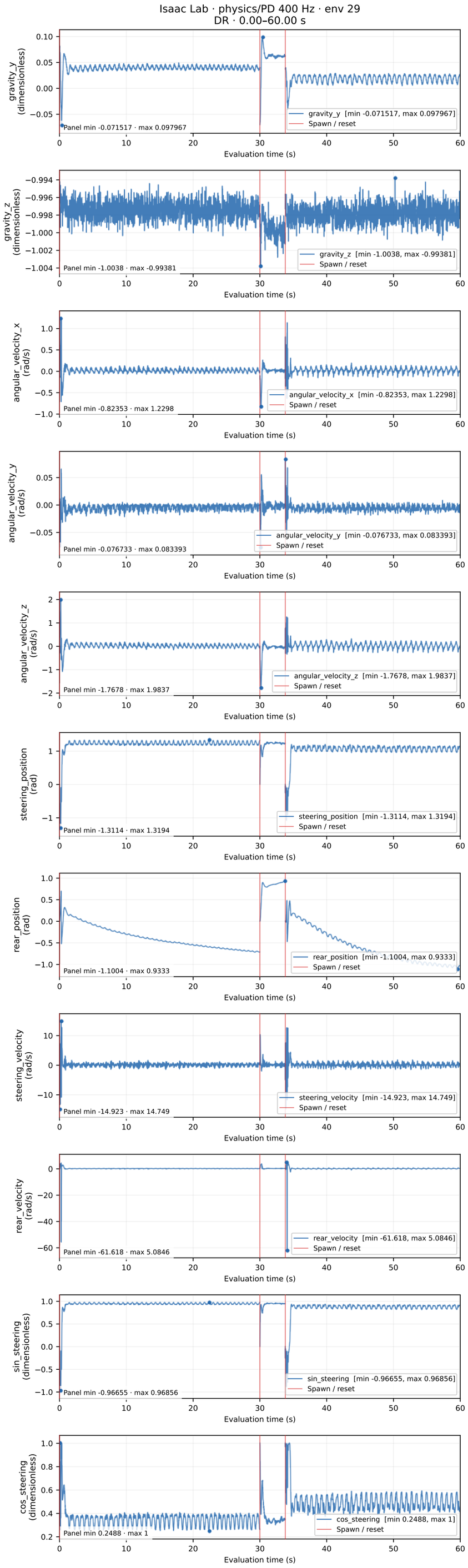
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 28; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10236, 1.12912, 1.19072, 1.15239]
  CoM XYZ mm [-0.636846, 3.37844, 3.57158]
  mu_s [0.579974, 0.673378, 0.713129, 0.785466]
  mu_d [0.579974, 0.673378, 0.713129, 0.52258]
  Kp [steer,rear] [10.2666, 0]; Kd [0.203469, 0]
  spawn roll 0.0384418 rad; rate 0.0541661 rad/s
  gyro bias XYZ rad/s [-0.000126931, -6.078e-05, 0.000257724]
  gravity bias YZ [0.00306666, -0.000674733]
E2 [30.00, 60.00) s:
  mass x [1.11974, 0.873552, 1.01957, 1.08793]
  CoM XYZ mm [-1.75007, -0.848766, 0.974908]
  mu_s [0.673378, 0.726175, 0.531594, 0.751971]
  mu_d [0.673378, 0.558574, 0.531594, 0.541224]
  Kp [steer,rear] [10.9783, 0]; Kd [0.200438, 0]
  spawn roll 0.0327392 rad; rate 0.00453961 rad/s
  gyro bias XYZ rad/s [0.000201461, 9.5479e-05, 3.07736e-05]
  gravity bias YZ [0.00220735, 0.00233325]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 28; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.952294, 1.08952, 0.928876, 1.18609]
  CoM XYZ mm [2.40347, -0.41802, -2.92448]
  mu_s [0.568073, 0.732187, 0.538434, 0.559972]
  mu_d [0.568073, 0.631664, 0.538434, 0.502209]
  Kp [steer,rear] [9.46634, 0]; Kd [0.193779, 0]
  spawn roll -0.0959019 rad; rate -0.208002 rad/s
  gyro bias XYZ rad/s [-8.36518e-05, 0.00020827, -0.000112668]
  gravity bias YZ [0.00304337, -0.00101055]
E2 [0.28, 30.28) s:
  mass x [0.93304, 1.00889, 1.11889, 1.08516]
  CoM XYZ mm [3.50917, 1.95555, 4.919]
  mu_s [0.631146, 0.528253, 0.700521, 0.732187]
  mu_d [0.631146, 0.528253, 0.641329, 0.631664]
  Kp [steer,rear] [10.2706, 0]; Kd [0.208555, 0]
  spawn roll 0.0571224 rad; rate -0.242375 rad/s
  gyro bias XYZ rad/s [0.000175128, -0.000326341, -0.000202794]
  gravity bias YZ [0.00267025, 0.000785849]
E3 [30.28, 30.66) s:
  mass x [1.04399, 0.869427, 0.823365, 0.934411]
  CoM XYZ mm [4.46926, -1.76103, -2.60666]
  mu_s [0.728342, 0.538434, 0.568073, 0.633024]
  mu_d [0.728342, 0.538434, 0.568073, 0.568172]
  Kp [steer,rear] [10.8384, 0]; Kd [0.193374, 0]
  spawn roll 0.00969489 rad; rate -0.226411 rad/s
  gyro bias XYZ rad/s [0.00011068, 0.000119604, 0.000314661]
  gravity bias YZ [-0.00140637, -0.00214619]
E4 [30.66, 31.00) s:
  mass x [1.19968, 1.01752, 1.0602, 0.913877]
  CoM XYZ mm [-0.991653, -1.08678, 4.77046]
  mu_s [0.559972, 0.71155, 0.541939, 0.633024]
  mu_d [0.502209, 0.71155, 0.534359, 0.568172]
  Kp [steer,rear] [9.41789, 0]; Kd [0.218262, 0]
  spawn roll -0.0938252 rad; rate 0.043656 rad/s
  gyro bias XYZ rad/s [-8.90731e-05, -3.84677e-05, 0.000531459]
  gravity bias YZ [-0.000366339, -0.0024708]
E5 [31.00, 31.54) s:
  mass x [1.06646, 0.984685, 1.00894, 1.10895]
  CoM XYZ mm [-2.39553, 4.82211, 1.03344]
  mu_s [0.749678, 0.666375, 0.559972, 0.538976]
  mu_d [0.741429, 0.519145, 0.502209, 0.538976]
  Kp [steer,rear] [9.37931, 0]; Kd [0.191366, 0]
  spawn roll -0.0513507 rad; rate 0.0101969 rad/s
  gyro bias XYZ rad/s [-0.000205965, 2.3688e-05, 0.00025687]
  gravity bias YZ [-2.8465e-06, -0.000470191]
E6 [31.54, 32.68) s:
  mass x [0.972611, 0.832211, 0.87056, 1.18059]
  CoM XYZ mm [4.07078, -1.95287, 1.45444]
  mu_s [0.640016, 0.733515, 0.640016, 0.69316]
  mu_d [0.513141, 0.558392, 0.513141, 0.69316]
  Kp [steer,rear] [9.46859, 0]; Kd [0.201181, 0]
  spawn roll -0.0680625 rad; rate 0.236171 rad/s
  gyro bias XYZ rad/s [0.000200285, 0.000310343, 0.000162168]
  gravity bias YZ [-0.00164235, -0.00253692]
E7 [32.68, 34.08) s:
  mass x [0.972775, 1.04356, 1.14376, 0.884581]
  CoM XYZ mm [-0.413933, 0.0959689, 2.696]
  mu_s [0.704749, 0.640016, 0.71008, 0.616244]
  mu_d [0.541926, 0.513141, 0.59371, 0.586498]
  Kp [steer,rear] [9.20953, 0]; Kd [0.197529, 0]
  spawn roll -0.0263583 rad; rate -0.0092295 rad/s
  gyro bias XYZ rad/s [-0.000194922, -0.000416148, -0.000107549]
  gravity bias YZ [-0.00426185, 3.11676e-05]
E8 [34.08, 34.50) s:
  mass x [1.18303, 0.986336, 1.03512, 0.807493]
  CoM XYZ mm [-0.762478, 2.35782, -0.110923]
  mu_s [0.568073, 0.541939, 0.736077, 0.69316]
  mu_d [0.568073, 0.534359, 0.699455, 0.69316]
  Kp [steer,rear] [10.2904, 0]; Kd [0.213506, 0]
  spawn roll -0.0150451 rad; rate -0.24368 rad/s
  gyro bias XYZ rad/s [0.000322825, -0.000121041, 4.10367e-05]
  gravity bias YZ [0.00235073, -0.00210662]
E9 [34.50, 54.24) s:
  mass x [1.06933, 1.11779, 0.925928, 0.881714]
  CoM XYZ mm [2.58632, -0.582327, -4.11222]
  mu_s [0.616244, 0.732187, 0.538976, 0.538976]
  mu_d [0.586498, 0.631664, 0.538976, 0.538976]
  Kp [steer,rear] [10.128, 0]; Kd [0.182089, 0]
  spawn roll -0.0443302 rad; rate -0.0679515 rad/s
  gyro bias XYZ rad/s [-0.000133908, -0.000311232, 2.11184e-05]
  gravity bias YZ [0.00115997, 0.000949268]
E10 [54.24, 60.00) s:
  mass x [0.857086, 1.16586, 1.1047, 0.989514]
  CoM XYZ mm [-1.1716, 4.02087, -3.10204]
  mu_s [0.727426, 0.71008, 0.791209, 0.640867]
  mu_d [0.606358, 0.59371, 0.767936, 0.556841]
  Kp [steer,rear] [9.13905, 0]; Kd [0.208661, 0]
  spawn roll 0.0237928 rad; rate -0.110249 rad/s
  gyro bias XYZ rad/s [0.000181704, -5.97139e-06, -3.66438e-05]
  gravity bias YZ [0.00189326, -0.000104177]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 28; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.58) s:
  mass x [0.952294, 1.08952, 0.928876, 1.18609]
  CoM XYZ mm [2.40347, -0.41802, -2.92448]
  mu_s [0.568073, 0.732187, 0.538434, 0.559972]
  mu_d [0.568073, 0.631664, 0.538434, 0.502209]
  Kp [steer,rear] [9.46634, 0]; Kd [0.193779, 0]
  spawn roll -0.0959019 rad; rate -0.208002 rad/s
  gyro bias XYZ rad/s [-8.36518e-05, 0.00020827, -0.000112668]
  gravity bias YZ [0.00304337, -0.00101055]
E2 [0.58, 30.58) s:
  mass x [0.937336, 0.889866, 1.1976, 0.989658]
  CoM XYZ mm [-0.185031, 1.64127, -0.30178]
  mu_s [0.637675, 0.723429, 0.757579, 0.757579]
  mu_d [0.637675, 0.723429, 0.70921, 0.70921]
  Kp [steer,rear] [10.4582, 0]; Kd [0.215546, 0]
  spawn roll -0.00614233 rad; rate 0.123039 rad/s
  gyro bias XYZ rad/s [4.75919e-05, -1.56271e-05, -0.000574203]
  gravity bias YZ [-0.00245388, -0.00177541]
E3 [30.58, 60.00) s:
  mass x [1.01946, 0.881359, 1.05001, 0.98384]
  CoM XYZ mm [1.85371, 2.16592, -3.81601]
  mu_s [0.723429, 0.749678, 0.736077, 0.727426]
  mu_d [0.723429, 0.741429, 0.699455, 0.606358]
  Kp [steer,rear] [9.88319, 0]; Kd [0.207719, 0]
  spawn roll 0.0637601 rad; rate -0.196215 rad/s
  gyro bias XYZ rad/s [5.14055e-06, -0.000353386, -0.000115272]
  gravity bias YZ [-0.000504244, -0.000612158]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 29 · full 60 s



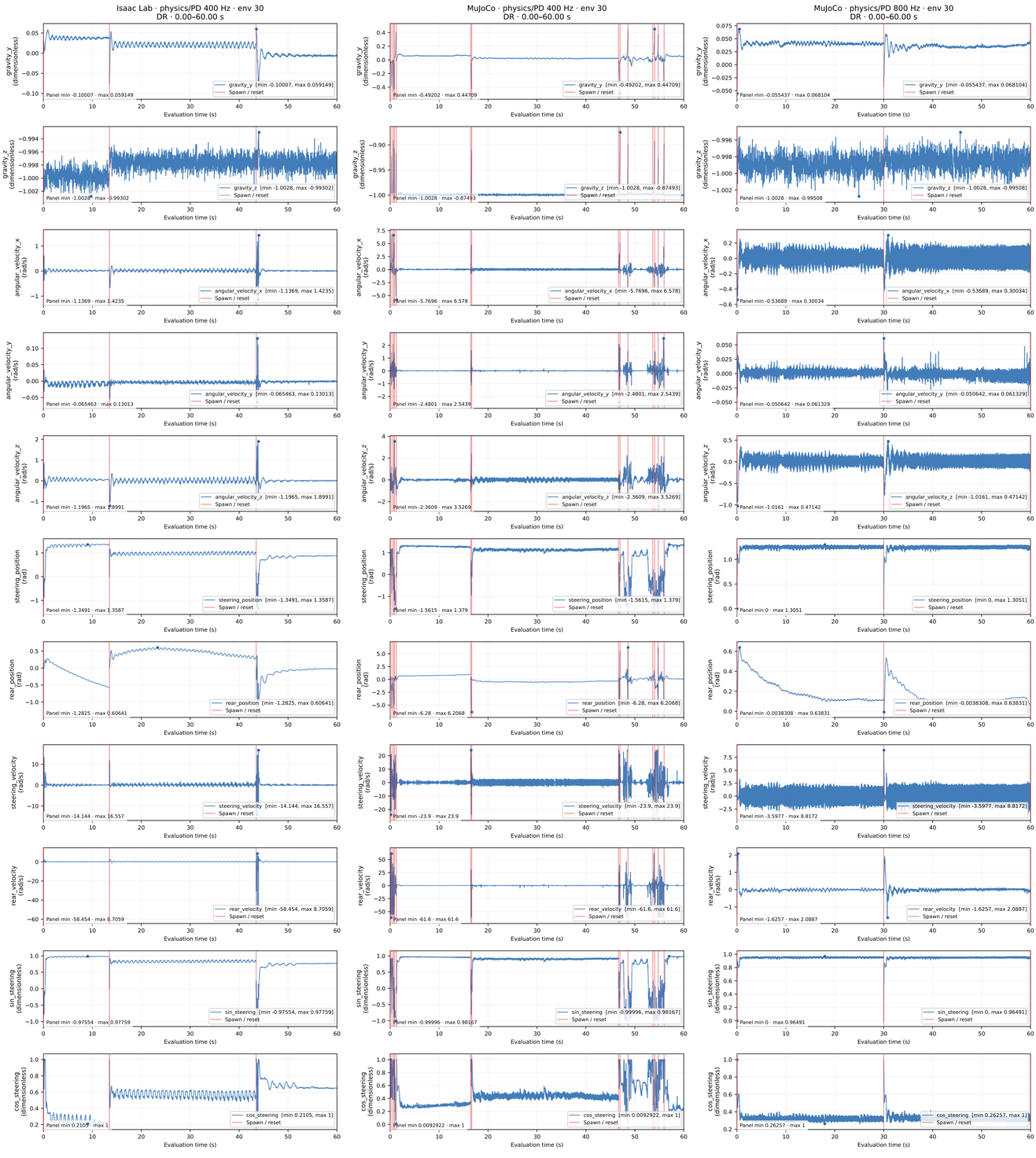
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 29; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04223, 0.986278, 1.12749, 1.0329]
  CoM XYZ mm [4.67661, -1.2452, 4.37889]
  mu_s [0.741039, 0.539956, 0.673378, 0.751971]
  mu_d [0.614405, 0.539956, 0.673378, 0.541224]
  Kp [steer,rear] [10.2652, 0]; Kd [0.185641, 0]
  spawn roll -0.0778912 rad; rate -0.12325 rad/s
  gyro bias XYZ rad/s [-0.000114268, 0.000298355, 0.00044557]
  gravity bias YZ [0.00208657, 0.00182313]
E2 [30.00, 33.80) s:
  mass x [1.08136, 1.17486, 1.0021, 0.847269]
  CoM XYZ mm [-0.536606, -4.60852, 4.68202]
  mu_s [0.726175, 0.560535, 0.531594, 0.531594]
  mu_d [0.558574, 0.560535, 0.531594, 0.531594]
  Kp [steer,rear] [10.49, 0]; Kd [0.203965, 0]
  spawn roll 0.0716322 rad; rate -0.183393 rad/s
  gyro bias XYZ rad/s [0.000145528, 0.000137405, -0.000204785]
  gravity bias YZ [0.000279636, -0.00157764]
E3 [33.80, 60.00) s:
  mass x [1.06325, 1.00468, 0.822012, 1.06386]
  CoM XYZ mm [-3.90828, 0.396512, -4.04747]
  mu_s [0.751971, 0.751971, 0.547743, 0.787792]
  mu_d [0.541224, 0.541224, 0.547743, 0.617134]
  Kp [steer,rear] [10.9457, 0]; Kd [0.214372, 0]
  spawn roll -0.0390346 rad; rate -0.167397 rad/s
  gyro bias XYZ rad/s [0.000233973, -2.20054e-05, -0.000181076]
  gravity bias YZ [-0.00162685, 0.00190149]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 29; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.901365, 1.01102, 1.10439, 1.0539]
  CoM XYZ mm [-2.95828, -4.5423, 0.231654]
  mu_s [0.723429, 0.69316, 0.633024, 0.538976]
  mu_d [0.723429, 0.69316, 0.568172, 0.538976]
  Kp [steer,rear] [9.07765, 0]; Kd [0.195074, 0]
  spawn roll -0.0440389 rad; rate -0.170196 rad/s
  gyro bias XYZ rad/s [7.55189e-05, 6.51789e-05, -0.000228312]
  gravity bias YZ [0.00108292, -0.000849417]
E2 [30.00, 30.26) s:
  mass x [1.15712, 1.14701, 0.925754, 1.00218]
  CoM XYZ mm [0.654529, 3.45013, 4.30141]
  mu_s [0.538434, 0.728342, 0.546287, 0.538976]
  mu_d [0.538434, 0.728342, 0.546287, 0.538976]
  Kp [steer,rear] [10.7335, 0]; Kd [0.186154, 0]
  spawn roll -0.0988358 rad; rate 0.0124952 rad/s
  gyro bias XYZ rad/s [0.000154901, 0.000115283, -5.695e-05]
  gravity bias YZ [-0.00325704, 5.60172e-05]
E3 [30.26, 31.56) s:
  mass x [1.12305, 1.17015, 1.15517, 0.865258]
  CoM XYZ mm [-1.57713, -0.299899, 0.111842]
  mu_s [0.749678, 0.791209, 0.669571, 0.728342]
  mu_d [0.741429, 0.767936, 0.669571, 0.728342]
  Kp [steer,rear] [10.6883, 0]; Kd [0.18963, 0]
  spawn roll -0.00759973 rad; rate -0.150361 rad/s
  gyro bias XYZ rad/s [0.000016499, -0.000012605, 0.000268673]
  gravity bias YZ [0.00414226, 0.00181435]
E4 [31.56, 33.36) s:
  mass x [0.994299, 0.90087, 1.0446, 1.10651]
  CoM XYZ mm [-4.55593, 0.216405, -0.69874]
  mu_s [0.541939, 0.727426, 0.631146, 0.528253]
  mu_d [0.534359, 0.606358, 0.631146, 0.528253]
  Kp [steer,rear] [9.10137, 0]; Kd [0.216598, 0]
  spawn roll -0.0415815 rad; rate -0.00592296 rad/s
  gyro bias XYZ rad/s [0.000175236, 0.000150931, 0.000124915]
  gravity bias YZ [0.000199283, -0.00204661]
E5 [33.36, 34.46) s:
  mass x [0.81369, 0.857251, 1.10322, 1.14334]
  CoM XYZ mm [0.838735, 3.11839, -3.82533]
  mu_s [0.541939, 0.538976, 0.728342, 0.640867]
  mu_d [0.534359, 0.538976, 0.728342, 0.556841]
  Kp [steer,rear] [9.12319, 0]; Kd [0.201365, 0]
  spawn roll -0.0193479 rad; rate 0.139358 rad/s
  gyro bias XYZ rad/s [0.000185854, -5.29748e-05, 7.57583e-05]
  gravity bias YZ [-0.00245777, 0.00206656]
E6 [34.46, 35.40) s:
  mass x [1.14994, 1.0652, 0.887675, 0.832326]
  CoM XYZ mm [4.13229, 4.01106, -0.34915]
  mu_s [0.757579, 0.546287, 0.69316, 0.631146]
  mu_d [0.70921, 0.546287, 0.69316, 0.631146]
  Kp [steer,rear] [10.9448, 0]; Kd [0.205048, 0]
  spawn roll 0.0030687 rad; rate -0.118402 rad/s
  gyro bias XYZ rad/s [7.88943e-05, -0.000186644, 0.000384477]
  gravity bias YZ [-0.00249443, 0.00376283]
E7 [35.40, 60.00) s:
  mass x [1.17704, 0.998073, 0.97533, 0.980104]
  CoM XYZ mm [-0.150428, 3.38569, -3.16444]
  mu_s [0.791209, 0.637675, 0.538976, 0.633024]
  mu_d [0.767936, 0.637675, 0.538976, 0.568172]
  Kp [steer,rear] [10.07, 0]; Kd [0.180639, 0]
  spawn roll -0.0282665 rad; rate -0.128308 rad/s
  gyro bias XYZ rad/s [9.44895e-05, -8.33654e-05, -6.65546e-05]
  gravity bias YZ [8.47903e-05, -0.000443589]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 29; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.901365, 1.01102, 1.10439, 1.0539]
  CoM XYZ mm [-2.95828, -4.5423, 0.231654]
  mu_s [0.723429, 0.69316, 0.633024, 0.538976]
  mu_d [0.723429, 0.69316, 0.568172, 0.538976]
  Kp [steer,rear] [9.07765, 0]; Kd [0.195074, 0]
  spawn roll -0.0440389 rad; rate -0.170196 rad/s
  gyro bias XYZ rad/s [7.55189e-05, 6.51789e-05, -0.000228312]
  gravity bias YZ [0.00108292, -0.000849417]
E2 [30.00, 30.28) s:
  mass x [1.15712, 1.14701, 0.925754, 1.00218]
  CoM XYZ mm [0.654529, 3.45013, 4.30141]
  mu_s [0.538434, 0.728342, 0.546287, 0.538976]
  mu_d [0.538434, 0.728342, 0.546287, 0.538976]
  Kp [steer,rear] [10.7335, 0]; Kd [0.186154, 0]
  spawn roll -0.0988358 rad; rate 0.0124952 rad/s
  gyro bias XYZ rad/s [0.000154901, 0.000115283, -5.695e-05]
  gravity bias YZ [-0.00325704, 5.60172e-05]
E3 [30.28, 30.92) s:
  mass x [0.896302, 0.960689, 1.12332, 0.967739]
  CoM XYZ mm [-2.92105, -2.56118, 1.92169]
  mu_s [0.791209, 0.546287, 0.528253, 0.69316]
  mu_d [0.767936, 0.546287, 0.528253, 0.69316]
  Kp [steer,rear] [10.9657, 0]; Kd [0.201706, 0]
  spawn roll -0.0789953 rad; rate 0.205389 rad/s
  gyro bias XYZ rad/s [-8.82892e-05, -9.22037e-05, -3.24904e-05]
  gravity bias YZ [-0.000913649, 0.00295795]
E4 [30.92, 32.88) s:
  mass x [1.19163, 0.952337, 0.964596, 0.953458]
  CoM XYZ mm [-4.88524, 2.50624, 0.287979]
  mu_s [0.611239, 0.71155, 0.791209, 0.616244]
  mu_d [0.611239, 0.71155, 0.767936, 0.586498]
  Kp [steer,rear] [10.1973, 0]; Kd [0.21061, 0]
  spawn roll -0.0341768 rad; rate -0.0950615 rad/s
  gyro bias XYZ rad/s [-4.88644e-05, 8.31145e-05, -0.000226066]
  gravity bias YZ [0.000706437, 0.000516435]
E5 [32.88, 60.00) s:
  mass x [1.04432, 1.13414, 1.1779, 1.14262]
  CoM XYZ mm [4.7274, -1.36799, -0.577485]
  mu_s [0.748289, 0.728342, 0.748289, 0.616244]
  mu_d [0.689499, 0.728342, 0.689499, 0.586498]
  Kp [steer,rear] [9.3829, 0]; Kd [0.196605, 0]
  spawn roll -0.0652417 rad; rate 0.233495 rad/s
  gyro bias XYZ rad/s [-0.000249581, 6.77096e-05, 4.06295e-05]
  gravity bias YZ [0.000499155, -0.00600322]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 30 · full 60 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 30; body order: base, rear_wheel_l, steering_l, front_wheel_l
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 13.48) s:
   mass x [0.734668, 1.07947, 1.12819, 1.14008]
   CoM XYZ mm [-3.09583, -2.85516, 0.8478311]
   mu_s [0.579974, 0.703711, 0.693205, 0.598343]
   mu_d [0.579974, 0.703711, 0.693205, 0.598343]
   Kp [steer,rear] [18.1628, 0]; Kd [0.185653, 0]
   spawn roll -0.0300911 rad; rate -0.232015 rad/s
   gyro bias XYZ rad/s [0.0092150005, 0.00012291, -0.000360708]
   gravity bias YZ [-0.0029152, -0.0005813311]
E2 [13.48, 43.48) s:
   mass x [0.997325, 1.1252, 1.01393, 0.844822]
   CoM XYZ mm [4.16046, -1.14143, 4.34159]
   mu_s [0.628821, 0.751971, 0.741039, 0.785466]
   mu_d [0.628821, 0.541224, 0.6144405, 0.52258]
   Kp [steer,rear] [9.06119, 0]; Kd [0.18606, 0]
   spawn roll 0.0747708 rad; rate 0.100169 rad/s
   gyro bias XYZ rad/s [-0.000290741, 4.18535e-05, 0.000158674]
   gravity bias YZ [-0.000980066, 0.002178981]
E3 [43.48, 60.00) s:
   mass x [0.831246, 0.886655, 0.923206, 0.906771]
   CoM XYZ mm [2.95128, -4.11288, 2.912911]
   mu_s [0.547743, 0.532645, 0.794338, 0.592046]
   mu_d [0.547743, 0.532645, 0.526209, 0.534947]
   Kp [steer,rear] [10.178, 0]; Kd [0.193007, 0]
   spawn roll -0.0606137 rad; rate 0.22692 rad/s
   gyro bias XYZ rad/s [6.64626e-06, -4.15746e-06, 1.02434e-05]
   gravity bias YZ [0.000547827, 0.00211663]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 30; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.34) s:
   mass x [1.13533, 0.80267, 0.892579, 0.845694]
   CoM XYZ mm [2.15583, -1.89545, 0.7330787]
   mu_s [0.559972, 0.757579, 0.640867, 0.748289]
   mu_d [0.502209, 0.70921, 0.556841, 0.6894999]
   Kp [steer,rear] [10.9072, 0]; Kd [0.193931, 0]
   spawn roll 0.0574297 rad; rate -0.165354 rad/s
   gyro bias XYZ rad/s [0.00015069, -1.76586e-05, -0.000395626]
   gravity bias YZ [0.000517843, 0.000246497]
E2 [0.34, 0.68) s:
   mass x [0.814908, 0.930503, 1.00509, 1.18996]
   CoM XYZ mm [-4.90881, -4.36729, -1.882954]
   mu_s [0.710108, 0.669571, 0.748289, 0.5282553]
   mu_d [0.59371, 0.6699571, 0.6894999, 0.5282553]
   Kp [steer,rear] [10.8369, 0]; Kd [0.208819, 0]
   spawn roll 0.0581603 rad; rate -0.137526 rad/s
   gyro bias XYZ rad/s [0.00016799, 2.68466e-05, 8.12228e-05]
   gravity bias YZ [-0.00577595, -0.00572445]
E3 [0.68, 0.98) s:
   mass x [1.19582, 0.826954, 0.940737]
   CoM XYZ mm [2.94838, -4.74771, 2.048791]
   mu_s [0.631146, 0.541939, 0.543359, 0.7047749]
   mu_d [0.631146, 0.534359, 0.534359, 0.5419926]
   Kp [steer,rear] [10.2526, 0]; Kd [0.21078, 0]
   spawn roll 0.0200943 rad; rate 0.184764 rad/s
   gyro bias XYZ rad/s [-0.00015187, 9.23402e-05, 0.000155635]
   gravity bias YZ [0.000014534, -0.000752591]
E4 [0.98, 1.26) s:
   mass x [1.0366, 1.17037, 0.903821, 0.866983]
   CoM XYZ mm [-3.62126, -4.25611, -1.90264]
   mu_s [0.71008, 0.541939, 0.728342, 0.7047749]
   mu_d [0.59371, 0.534359, 0.728342, 0.541926]
   Kp [steer,rear] [10.7814, 0]; Kd [0.202091, 0]
   spawn roll -0.0386995 rad; rate -0.137435 rad/s
   gyro bias XYZ rad/s [0.000173203, -2.58349e-06, 0.000304482]
   gravity bias YZ [-0.000570498, -0.00117011]
E5 [1.26, 16.46) s:
   mass x [0.942509, 0.814669, 0.926681, 1.19858]
   CoM XYZ mm [-1.89303, -4.76264, -0.8495811]
   mu_s [0.538434, 0.643024, 0.69316, 0.757579]
   mu_d [0.538434, 0.560172, 0.69316, 0.70921]
   Kp [steer,rear] [9.83469, 0]; Kd [0.211493, 0]
   spawn roll 0.0100802 rad; rate -0.218889 rad/s
   gyro bias XYZ rad/s [0.000160507, -0.000274877, -0.00013085]
   gravity bias YZ [0.00150107, -0.00064928]
E6 [16.46, 16.66) s:
   mass x [0.946566, 1.17225, 1.09063, 0.804103]
   CoM XYZ mm [-0.174554, 0.08833971, 1.76314]
   mu_s [0.728342, 0.736077, 0.546287, 0.6112239]
   mu_d [0.728342, 0.699455, 0.546287, 0.6112239]
   Kp [steer,rear] [10.9993, 0]; Kd [0.195934, 0]
   spawn roll 0.0770678 rad; rate -0.00949378 rad/s
   gyro bias XYZ rad/s [-4.24715e-05, -6.68252e-05, -0.000254964]
   gravity bias YZ [-0.00110171, -0.00164844]
E7 [16.66, 48.66) s:
   mass x [0.810637, 1.03981, 1.14723, 1.1206]
   CoM XYZ mm [2.22689, 0.925924, -4.883461]
   mu_s [0.757579, 0.538434, 0.748289, 0.5599072]
   mu_d [0.70921, 0.538434, 0.689499, 0.5022099]
   Kp [steer,rear] [9.69393, 0]; Kd [0.214131, 0]
   spawn roll 0.0192156 rad; rate -0.227771 rad/s
   gyro bias XYZ rad/s [-9.3951e-05, -3.74562e-05, 0.000290037]
   gravity bias YZ [0.00207558, 0.000515313]
E8 [48.66, 96.86) s:
   mass x [0.91235, 1.1217, 1.09581, 0.858619]
   CoM XYZ mm [-1.508392, 4.804178, -0.6672561]
   mu_s [0.757579, 0.736077, 0.749678, 0.728342]
   mu_d [0.70921, 0.699455, 0.741429, 0.728342]
   Kp [steer,rear] [10.1318, 0]; Kd [0.200717, 0]
   spawn roll 0.0444284 rad; rate 0.0918835 rad/s
   gyro bias XYZ rad/s [0.000170838, -0.000171349, -3.40807e-06]
   gravity bias YZ [-0.0009061515, -0.00179085]
E9 [48.96, 48.60) s:
   mass x [1.1734, 1.15025, 0.876941, 0.9934433]
   CoM XYZ mm [4.87098, 1.67359, -2.34641]
   mu_s [0.71008, 0.597748, 0.757579, 0.7482089]
   mu_d [0.59371, 0.597748, 0.70921, 0.689499]
   Kp [steer,rear] [10.3798, 0]; Kd [0.213412, 0]
   spawn roll 0.0402838 rad; rate 0.158738 rad/s
   gyro bias XYZ rad/s [0.000146746, -0.000149573, -5.23109e-05]
   gravity bias YZ [0.000241813, 0.000491757]
E10 [48.60, 53.01) s:
   mass x [0.812724, 1.099431, 0.929454, 1.03354]
   CoM XYZ mm [-0.76654, 2.52297, -1.23623]
   mu_s [0.748289, 0.715515, 0.723429, 0.710081]
   mu_d [0.689499, 0.558392, 0.723429, 0.59371]
   Kp [steer,rear] [9.70706, 0]; Kd [0.193021, 0]
   spawn roll -0.0390949 rad; rate 0.215487 rad/s
   gyro bias XYZ rad/s [6.37738e-05, 0.00011428, 9.97606e-05]
   gravity bias YZ [0.00241983, 0.001947081
E11 [53.68, 54.04) s:
   mass x [1.0475, 1.08955, 0.975661, 1.13058]
   CoM XYZ mm [2.08039, 4.20953, -1.9372]
   mu_s [0.611239, 0.541939, 0.568073, 0.611239]
   mu_d [0.611239, 0.534359, 0.568073, 0.611239]
   Kp [steer,rear] [9.0622, 0]; Kd [0.209563, 0]
   spawn roll -0.0816654 rad; rate 0.226339 rad/s
   gyro bias XYZ rad/s [0.00018062, 0.000208024, -0.000175451]
   gravity bias YZ [0.0003351151, 0.000128454]
E12 [54.04, 54.68) s:
   mass x [0.867411, 1.11888, 0.961108, 0.899199]
   CoM XYZ mm [-4.60154, -1.61398, 3.00352]
   mu_s [0.727428, 0.710082, 0.71155, 0.732187]
   mu_d [0.606354, 0.59371, 0.71155, 0.631664]
   Kp [steer,rear] [10.2915, 0]; Kd [0.210856, 0]
   spawn roll -0.0422033 rad; rate 0.0399885 rad/s
   gyro bias XYZ rad/s [0.000379203, 0.000219853, 0.000175484]
   gravity bias YZ [0.00150931, 0.0008223312]
E13 [54.78, 55.98) s:
   mass x [0.7112, 1.06309, 0.882956, 1.04843]
   CoM XYZ mm [-1.124661, 0.67816, 3.7155]
   mu_s [0.723429, 0.749678, 0.71155, 0.71155]
   mu_d [0.723429, 0.741429, 0.71155, 0.71155]
   Kp [steer,rear] [9.38777, 0]; Kd [0.207596, 0]
   spawn roll -0.0713152 rad; rate -0.0630836 rad/s
   gyro bias XYZ rad/s [-1.12214e-05, 5.21473e-05, -0.000534602]
   gravity bias YZ [0.000144944, 0.000659295]
E14 [55.98, 60.00) s:
   mass x [0.800622, 1.15836, 1.0868, 0.876823]
   CoM XYZ mm [-1.475, 1.53447, -1.20504]
   mu_s [0.700521, 0.640016, 0.546287, 0.723429]
   mu_d [0.641329, 0.513141, 0.546287, 0.723429]
   Kp [steer,rear] [9.76497, 0]; Kd [0.19418, 0]
   spawn roll 0.0248617 rad; rate -0.14799 rad/s
   gyro bias XYZ rad/s [-0.000122648, 6.29751e-05, 0.000443132]
   gravity bias YZ [0.00233486, -0.00106821]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 30; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
   mass x [1.13533, 0.80267, 0.892579, 0.845694]
   CoM XYZ mm [2.15583, -1.89545, -0.7330787]
   mu_s [0.559972, 0.757579, 0.640867, 0.748289]
   mu_d [0.502209, 0.70921, 0.556841, 0.6894999]
   Kp [steer,rear] [10.9072, 0]; Kd [0.193931, 0]
   spawn roll 0.0574297 rad; rate -0.165354 rad/s
   gyro bias XYZ rad/s [0.00015069, -1.76586e-05, -0.000395626]
   gravity bias YZ [0.000517843, 0.000246497]
E2 [30.00, 60.00) s:
   mass x [0.8566535, 1.04344, 1.00607, 1.18416]
   CoM XYZ mm [2.97977, -0.9424331, -4.27524]
   mu_s [0.791209, 0.71155, 0.704749, 0.597748]
   mu_d [0.767936, 0.71155, 0.541926, 0.597748]
   Kp [steer,rear] [9.25968, 0]; Kd [0.186814, 0]
   spawn roll 0.0439583 rad; rate -0.0288241 rad/s
   gyro bias XYZ rad/s [0.000535818, 0.000184314, 0.000122627]
   gravity bias YZ [-0.0008182223, 0.0008313351]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 31 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 31
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 31
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 31
DR · 0.00–60.00 s

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 31; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.888913, 0.920628, 1.16873, 0.999925]
  CoM XYZ mm [-2.61347, -1.67952, -0.69344]
  mu_s [0.547743, 0.787792, 0.559809, 0.673378]
  mu_d [0.547743, 0.617134, 0.559809, 0.673378]
  Kp [steer,rear] [10.5455, 0]; Kd [0.198474, 0]
  spawn roll -0.00862077 rad; rate 0.0478292 rad/s
  gyro bias XYZ rad/s [3.97985e-05, -0.000110208, -1.53622e-05]
  gravity bias YZ [0.000798645, 0.00177435]
E2 [30.00, 60.00) s:
  mass x [1.16364, 0.887446, 1.15393, 1.06103]
  CoM XYZ mm [-0.166044, -4.54881, -4.72042]
  mu_s [0.764681, 0.787792, 0.547743, 0.584424]
  mu_d [0.764681, 0.617134, 0.547743, 0.584424]
  Kp [steer,rear] [9.07667, 0]; Kd [0.215742, 0]
  spawn roll 0.0105771 rad; rate 0.105873 rad/s
  gyro bias XYZ rad/s [1.70873e-05, -9.48296e-05, -6.3715e-05]
  gravity bias YZ [1.16744e-05, -0.00196036]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 31; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.997054, 0.921347, 0.907427, 1.19406]
  CoM XYZ mm [-2.80109, -4.67992, 3.54697]
  mu_s [0.749678, 0.611239, 0.736077, 0.736077]
  mu_d [0.741429, 0.611239, 0.699455, 0.699455]
  Kp [steer,rear] [10.4465, 0]; Kd [0.184146, 0]
  spawn roll 0.0612062 rad; rate -0.192569 rad/s
  gyro bias XYZ rad/s [-0.000168277, 0.000281201, 0.000184886]
  gravity bias YZ [-0.000333241, 0.00361406]
E2 [30.00, 60.00) s:
  mass x [0.964597, 0.977022, 1.08373, 1.10108]
  CoM XYZ mm [2.24418, -0.39583, -3.62629]
  mu_s [0.748289, 0.71008, 0.791209, 0.597748]
  mu_d [0.689499, 0.59371, 0.767936, 0.597748]
  Kp [steer,rear] [9.78566, 0]; Kd [0.194591, 0]
  spawn roll -0.095277 rad; rate 0.228682 rad/s
  gyro bias XYZ rad/s [-1.55867e-05, -0.000193466, -4.36437e-05]
  gravity bias YZ [0.00434751, 0.0011049]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 31; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.997054, 0.921347, 0.907427, 1.19406]
  CoM XYZ mm [-2.80109, -4.67992, 3.54697]
  mu_s [0.749678, 0.611239, 0.736077, 0.736077]
  mu_d [0.741429, 0.611239, 0.699455, 0.699455]
  Kp [steer,rear] [10.4465, 0]; Kd [0.184146, 0]
  spawn roll 0.0612062 rad; rate -0.192569 rad/s
  gyro bias XYZ rad/s [-0.000168277, 0.000281201, 0.000184886]
  gravity bias YZ [-0.000333241, 0.00361406]
E2 [0.32, 0.72) s:
  mass x [0.966366, 1.19041, 0.936999, 1.10457]
  CoM XYZ mm [-4.39216, 2.49833, 3.69328]
  mu_s [0.749678, 0.704749, 0.640867, 0.69316]
  mu_d [0.741429, 0.541926, 0.556841, 0.69316]
  Kp [steer,rear] [10.7143, 0]; Kd [0.19111, 0]
  spawn roll -0.0976437 rad; rate -0.134834 rad/s
  gyro bias XYZ rad/s [0.000126475, -0.000147112, -0.00012958]
  gravity bias YZ [0.000159554, 0.00158182]
E3 [0.72, 30.72) s:
  mass x [0.900998, 0.800325, 1.1386, 1.04068]
  CoM XYZ mm [1.69883, 3.85388, 4.13658]
  mu_s [0.71155, 0.637675, 0.732187, 0.640867]
  mu_d [0.71155, 0.637675, 0.631664, 0.556841]
  Kp [steer,rear] [9.83752, 0]; Kd [0.215052, 0]
  spawn roll 0.0430423 rad; rate -0.109067 rad/s
  gyro bias XYZ rad/s [-0.000180353, 0.000203191, 0.00028134]
  gravity bias YZ [0.000324342, -0.000905106]
E4 [30.72, 60.00) s:
  mass x [0.957543, 0.910966, 0.804324, 0.859663]
  CoM XYZ mm [4.23565, 3.55379, -3.90607]
  mu_s [0.568073, 0.538976, 0.757579, 0.538976]
  mu_d [0.568073, 0.538976, 0.70921, 0.538976]
  Kp [steer,rear] [9.0268, 0]; Kd [0.216424, 0]
  spawn roll 0.0149589 rad; rate 0.184369 rad/s
  gyro bias XYZ rad/s [-0.000210905, -0.000187521, -0.000211918]
  gravity bias YZ [-0.00353316, -0.00337261]
Six significant digits shown; full precision: sampled_dr_values.json.
```